R-02:8 Arb. nr. 960

Final report:

Proposal number:	GRD1-1999-20007
Project acronym:	HAMSTAD
Project full title:	Determination of liquid water transfer properties of porous building materials and development of numerical assessment methods.

Chalmers University of Technology
December 2002

This report, the results of this work package and methodology of HAM-modeling is available on CD. The content of CD is presented in Appendix.

Summary

EU initiates this work in order to develop a platform for assessment of computational modeling of Heat, Moisture and Air (HAM) transport mechanism in building physics. The reference method for development of these new guidelines was Glaser method.

The results of this part of project contain a defined HAM mathematical simulation model and five benchmarks. These benchmarks could be used as references for assessment of existing and new software.

It should be mentioned that the benchmarks only cover one-dimensional cases. For accurate predictions of a buildings HAM-behavior two- and three-dimensional simulation models are required. This project group therefore suggests a continuation of this project in order to have assessment tools, which also covers two-dimensional models and benchmarks.

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1 Introduction

Building envelopes are simultaneously exposed to the indoor and outdoor climate. The differences in temperature, moisture and air pressure have a major impact on the sustainability of building envelopes. In order to improve sustainability of building components, the physical state and behavior of building envelope are issues of great interest and importance in civil engineering.

At present, the certified Glaser method for calculation, prediction and evaluation of moisture performance is considered as rarely applicable. The present state of knowledge and equipment concerning heat, air and moisture demands updating of standards and certifications. From this point of view the proposal of this project is to improve simulation models as much as possible.

1.1 Objectives

The objective of this part of project (WP2) is to propose a reference HAM-document, describing the physics of heat, moisture and air transfer including the treatment of the necessary material properties, boundary, initial and contact conditions. Five benchmark cases are defined and several numerical solutions for these are provided in this document. These benchmarks can be used as references in the CEN-assessment of soft ware packages.

1.2 Participants in this work group

Participants in this work package are:

- Chalmers University of Technology, Coordinator of WP' (CTH)
- TNO Building and Construction Research (TNO)
- University of Leuven (KUL)
- Technical University of Dresden (TUD)
- Technion Israel Institute of Technology (Technion)

In addition to these HAMSTAD participants additional numerical solutions has been provided by the following organizations and universities.

- Eindhoven University of Technology (TUE)
- National Research Council of Canada (NRC)
- Fraunhofer Institute of Building Physics (IBP)

1.3 Outline of the HAM-model

The simple Glaser model has been used in many countries to assess moisture safety of building components. This model has a lot of limitations. In order to improve the Glaser model, this document suggests simplifications in the physical models required in order to assess the HAM performance of structures.

The suggested model covers heat and mass balance, heat, moisture and air transfers, exterior- and interior boundary and climate condition. The model has some limitations, as examples:

- Temperature should be in the range of –30 °C to +80 °C.

- Effects associated with phase change, liquid to ice, are neglected.
- Climatic load due to driving rain is simplified.
- Sorption isotherm is temperature independent
- No hysteresis is accounted for
- No chemical reactions are considered.
- Ageing effects or changes in the geometrical dimensions are neglected

For a complete list of limitations, see the HAM-model reference document [1][1].

[1] Carl-Eric Hagentoft, 2002: HAMSTAD – WP2 Modeling, Report R-02:9. Gothenburg, Department of Building Physics, Chalmers University of Technology.

3. Benchmarks

Benchmarks presented in coming section covers heat, moisture and air transfer. Each benchmark covers at least two transfer mechanisms. The cases have been selected in order to cover various combinations of climatic loads and combinations of materials. The following list briefly describes the benchmarks and what is tested:

1. The first benchmark deals with internal condensation occurring at the contact surface between two materials. Materials have different thermal and moisture transport properties – one is capillary active load bearing material while the other is capillary non-active (infinite resistance to liquid flow) insulation; thermal conductivities of materials differ by factor 50 (at dry conditions).

2. The second benchmark deals with moisture redistribution in one material layer under isothermal conditions. Moisture movement is caused by the sudden changes in relative humidity of the surroundings. This case has analytical solution.

3. The third benchmark deals with an air transfer through one material layer. Moisture transfer is caused mainly by the air flow through the layer, and also by the moisture and temperature gradients across the layer.

4. The fourth benchmark deals with the moisture movement inside the wall with hygroscopic finishing material. The wall is submitted to subsequent relative humidity, heat and moisture loads at the inner and outer surface.

5. The fifth benchmark deals with the moisture redistribution inside the wall with capillary active inside insulation. The wall consists of three layers: load bearing, finishing and insulating material. Thermal conductivities of load bearing and insulating materials differ by factor 11 (at dry conditions).

3.1 Benchmark one: *'Insulated roof'*

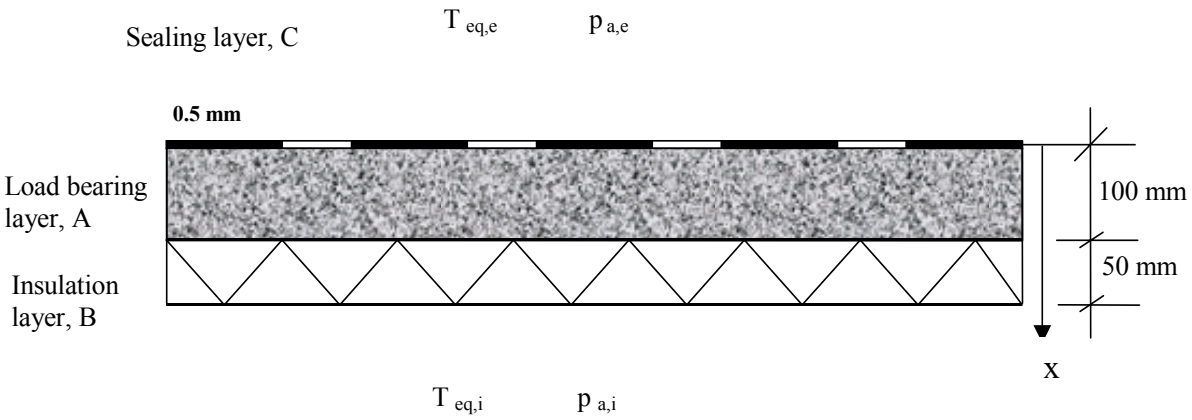
Insulated roof

HAMSTAD: Benchmark no: 1

Initiated by Chalmers University of Technology (CTH)

An insulated roof structure is analysed in one dimensions. The thermal insulation is facing the interior and there is a moisture barrier facing the exterior. The structure is perfectly airtight.

Structure



Properties of layers

1 General data:

$T = T_{ref} = 293.15$ K, ρ_w=1000 kg/m^3, R=8.314 J/(mol,K), M_w=0.018 kg/mol
ℓ_{lv} =2.5 · 10^6 J/kg

2 Material properties:

A. Load bearing material

-Water retention curve: $$w = \frac{146}{(1+(8\cdot 10^{-8}\cdot P_{suc})^{1.6})^{0.375}} \quad (\text{kg/m}^3)$$

$$P_{suc} = 0.125\cdot 10^8 ((\frac{146}{w})^{\frac{1}{0.375}} - 1)^{0.625} \quad (\text{Pa})$$

- Sorption isotherm: $$w = \frac{146}{(1+(-8\cdot 10^{-8}\cdot \frac{RT\rho_w}{M_w}\ln(\phi))^{1.6})^{0.375}} \quad (\text{kg/m}^3)$$

$$\phi = \exp(-\frac{M_w}{RT\rho_w} 0.125\cdot 10^8 ((\frac{146}{w})^{\frac{1}{0.375}} - 1)^{0.625}) \quad (-)$$

- Vapour diffusion: $$\delta_p = \frac{M_w}{RT}\cdot\frac{26.1\cdot 10^{-6}}{200}\cdot\frac{1-\frac{w}{146}}{0.503\cdot(1-\frac{w}{146})^2 + 0.497} \quad (\text{s})$$

- Liquid water permeability:

$$K = \exp(-39.2619 + 0.0704\cdot(w-73) - 1.7420\cdot 10^{-4}\cdot(w-73)^2 - 2.7953\cdot 10^{-6}\cdot(w-73)^3 - 1.1566\cdot 10^{-7}\cdot(w-73)^4 + 2.5969\cdot 10^{-9}\cdot(w-73)^5) \quad (\text{s})$$

- Thermal conductivity: $$\lambda = 1.5 + \frac{15.8}{1000} w \quad (\text{W/mK})$$

- Specific heat capacity: c_p=800 (J/kgK)

- Density: ρ_0= 2280 (kg/m^3)

B. Insulation material

-Water retention curve: $$w = \frac{900}{(1+(2\cdot 10^{-4}\cdot P_{suc})^2)^{0.5}} \quad (\text{kg/m}^3)$$

$$P_{suc} = 0.5 \cdot 10^4 ((\frac{900}{w})^2 - 1)^{0.5} \text{ (Pa)}$$

- Sorption isotherm: $$w = \frac{900}{(1 + (-2 \cdot 10^{-4} \cdot \frac{RT\rho_w}{M_w} \ln(\phi))^2)^{0.5}} \text{ (kg/m}^3\text{)}$$

$$\phi = \exp(-\frac{M_w}{RT\rho_w} 0.5 \cdot 10^4 ((\frac{900}{w})^2 - 1)^{0.5}) \text{ (-)}$$

- Vapour diffusion: $$\delta_p = \frac{M_w}{RT} \cdot \frac{26.1 \cdot 10^{-6}}{9.6} \cdot \frac{1 - \frac{w}{900}}{0.503 \cdot (1 - \frac{w}{900})^2 + 0.497} \text{ (s)}$$

- Liquid water permeability: $K = 0$ s

- Thermal conductivity: $\lambda = 0.033 + \frac{0.59}{1000} w$ (W/mK)

- Specific heat capacity: c_p=1000 (J/kgK)

- Density: ρ_0= 73.9 (kg/m^3)

C. Sealing layer

Z_p = 1 · 10^{12} m/s, K=0 s, R=0 m^{2}K/W, moisture and heat capacity negligible.

3 Dimensions:

A.	Load bearing layer:	0.1 m
B.	Insulation layer:	0.05 m
C.	Sealing layer:	0.0005 m

Boundary conditions

1 Thermal and humidity conditions:

According to attached climate data file: *ClimateBench1.txt.*
Hourly values covering one year. Interpolated values for intermediate times.

Repeated values for ensuing (subsequent) years shall be used.
Equivalent temperatures accounts for both ambient air temperature and radiation exchange.

Climate data are structured as:

Column 1	Column 2	Column 3	Column 4	Column5
time (s)	$T_{eq,e}$ (ºC)	$T_{eq,i}$ (ºC)	$p_{a,e}$ (Pa)	$p_{a,i}$ (Pa)

2 Pressure conditions: no air pressure difference

3 Surface transfer coefficients:

$\alpha_{e,e}$ = 25 W/m^{2}K, $\alpha_{e,i}$ =7 W/m^{2}K, $\beta_{p,e}$ = 0 s/m , $\beta_{p,i}$ =2 · 10^{-8} s/m

Initial conditions

A. Load bearing material: w=145 kg/m^3, T=10 ºC
B. Insulation layer material: w=0.065 kg/m^3, T=10 ºC

Output

Output file has ASCII format. Each line represents a given time (h). Data are separated by a space. Each year is represented by in total 8761 lines. The start of the year is 0 h and the last hour is 8760.

1. **Simulation time: 5 years**

2. **Requested output:**

Moisture content $w(x)$ (kg/m^3) in space and time for the load bearing material, A and the insulation material, B, at the depth x (m) from the top of the load bearing material.

w(0167) w(0.05) w(0.0834) w(0.125)

Output file structure: files no. 1-5: (one file for each year!), in ASCII format

Column 1	Column 2	Column 3	Column 4	Column 5
time	w(0.0167)	w(0.05)	w(0.0834)	w(0.125)
0	145	145	145	0.065
1	…	…	…	…
2	…	…	…	…
…	…	…	…	…
8760	…	…	…	…

Integrated mass of moisture M (kg/m^2) in each layer, and heat flow into the structure from the interior q (W/m^2).

M_A M_B q

Output file structure: files no. 6-10: (one file for each year!), in ASCII format

Column 1	Column 2	Column 3	Column 4
time	M_A	M_B	q
0	14.5	14.5	0.00325
1	…	…	…
2	…	…	…
…	…	…	…
8760	…	…	…

3 Name of data files:

The output files should be named as: 'University/Institute'+ Bench1+'Output file no.' + .txt, e.g. files from Chalmers (CTH):

CTHBench11.txt
CTHBench12.txt
…
CTHBench110.txt

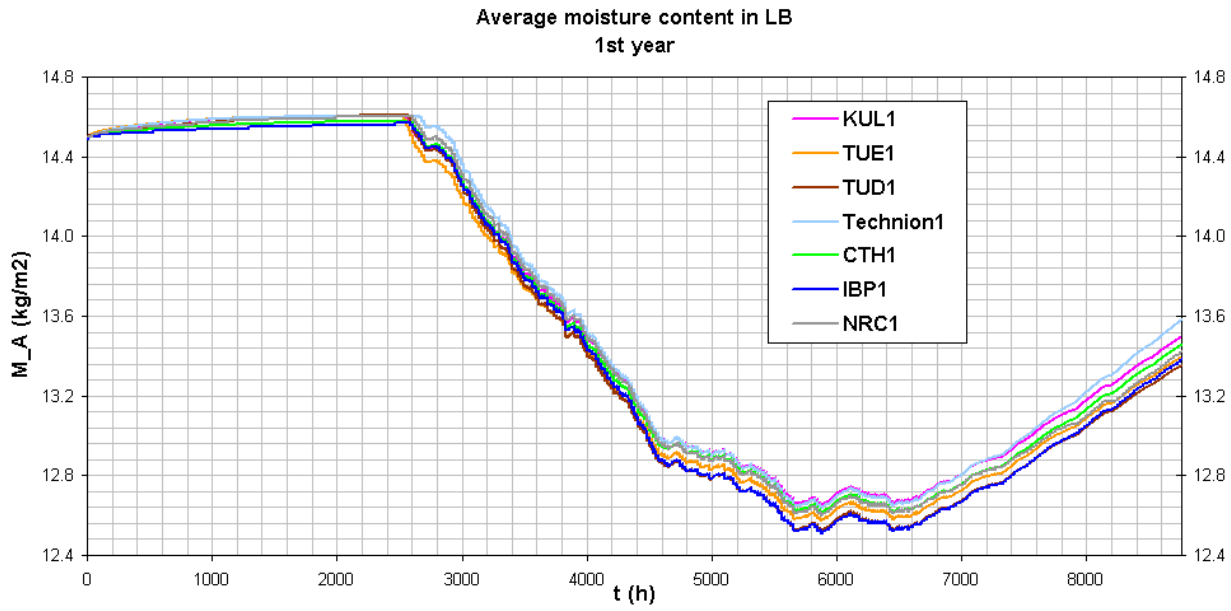


Figure 3.1.1

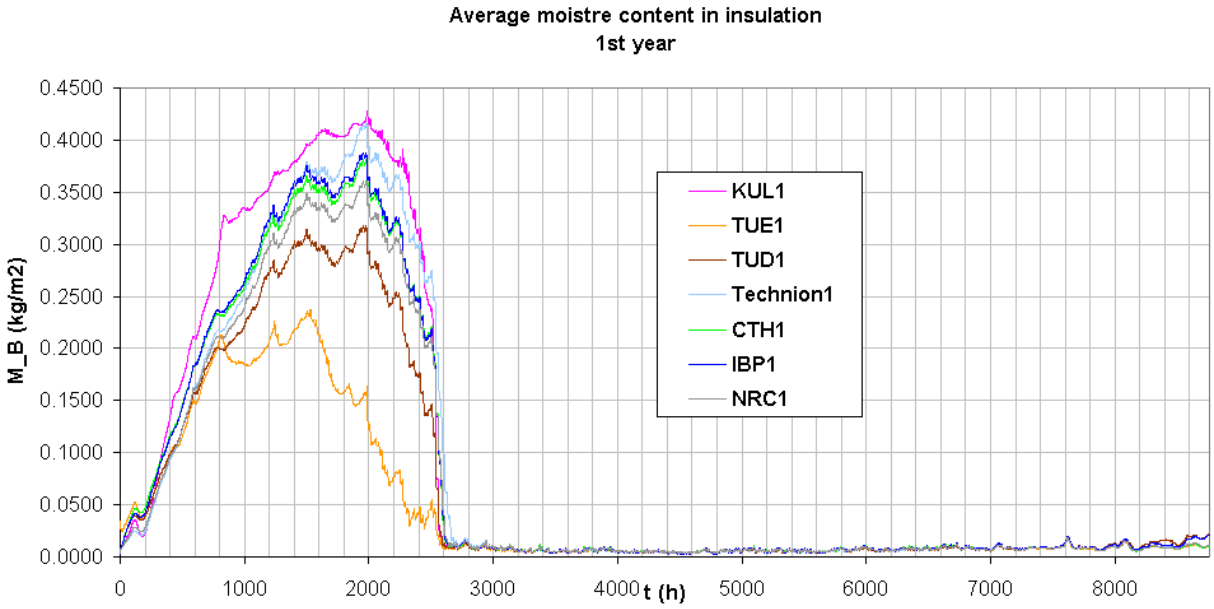


Figure 3.1.2

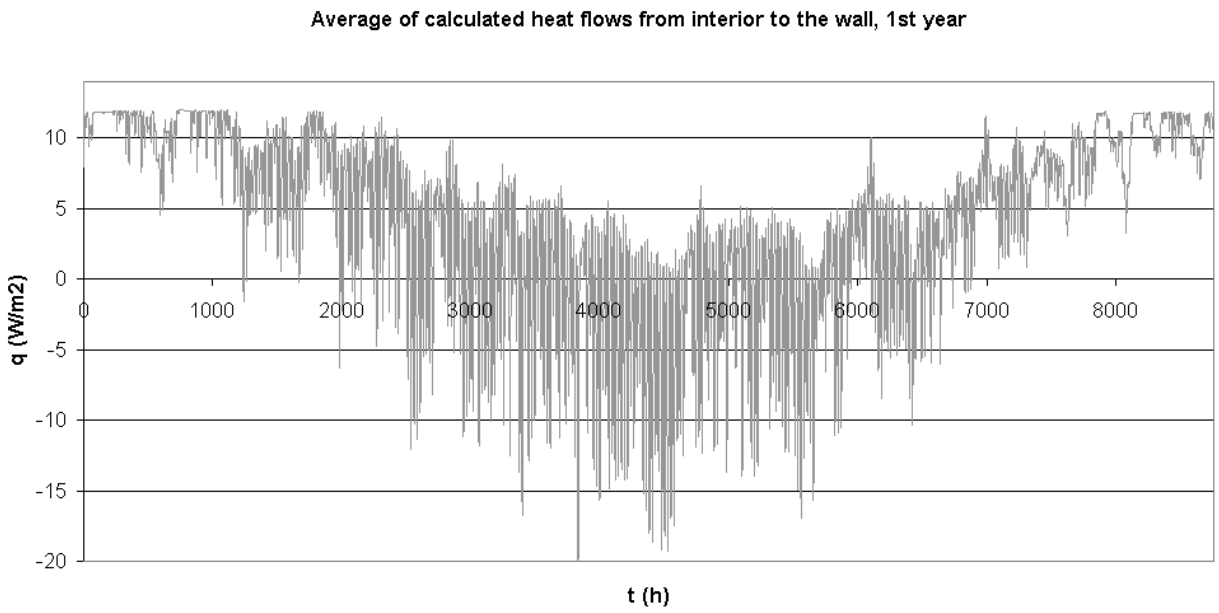


Figure 3.1.3

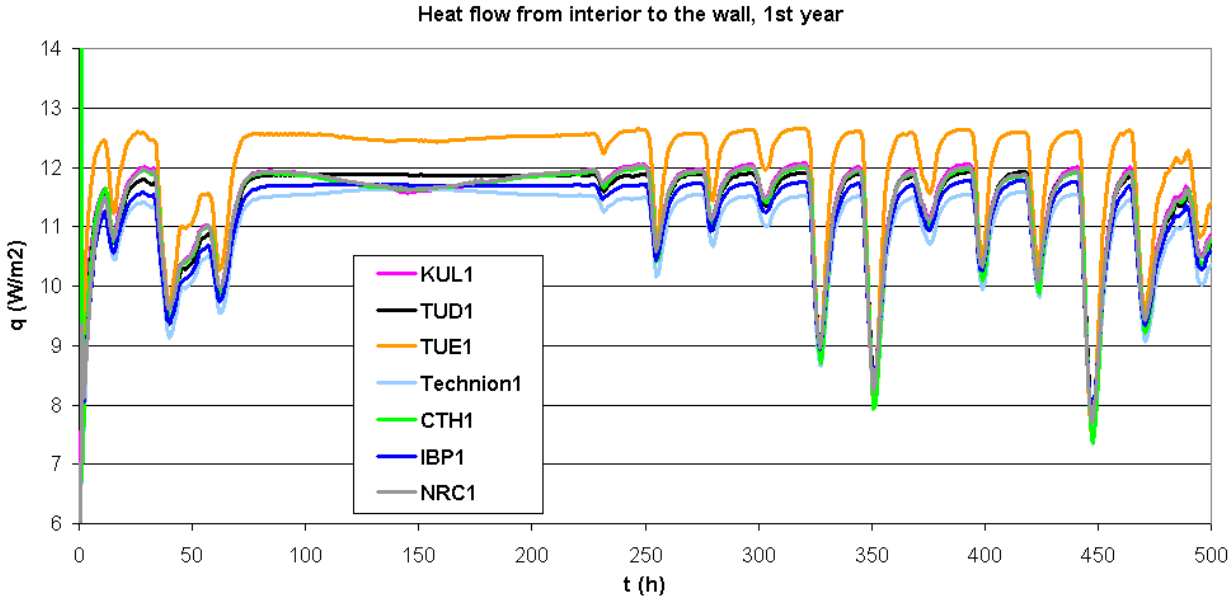


Figure 3.1.4

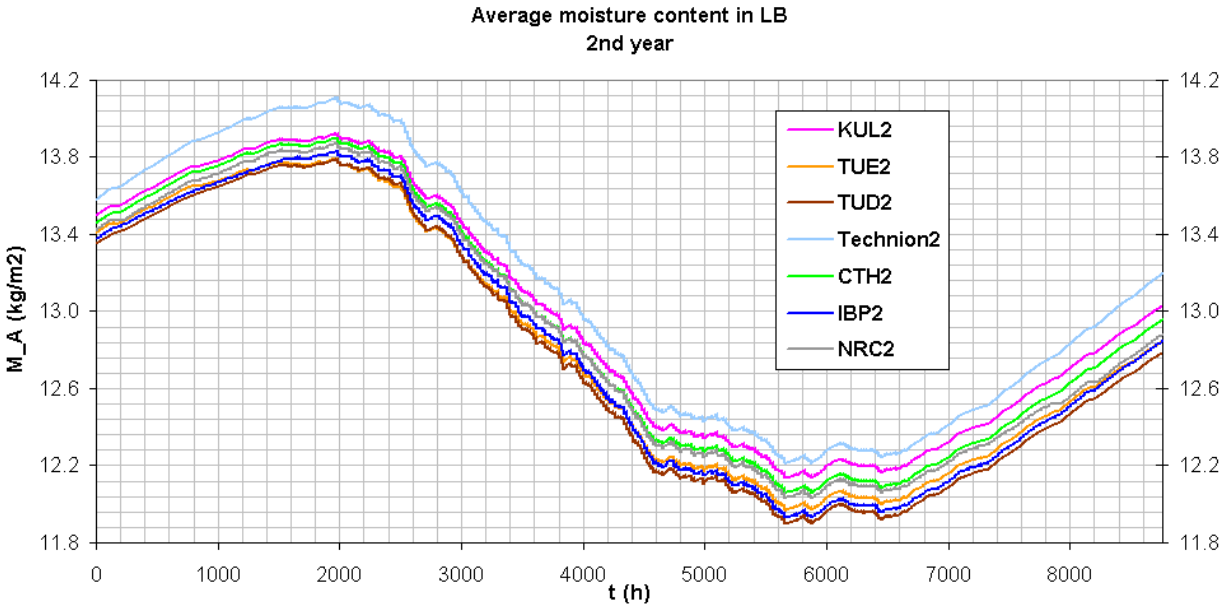


Figure 3.1.5

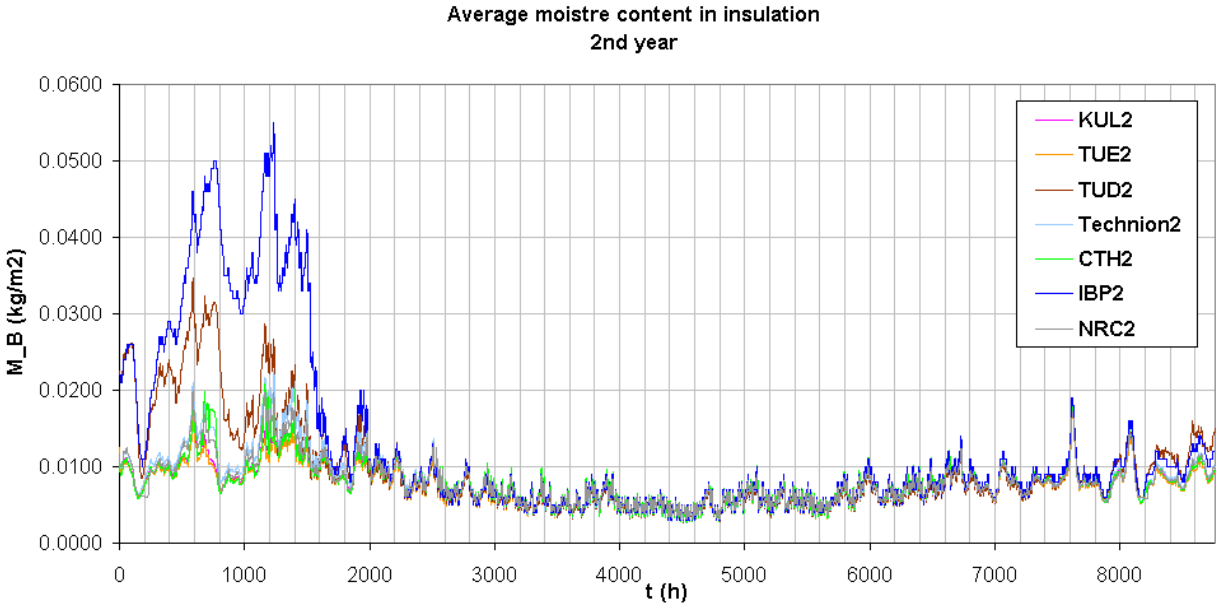


Figure 3.1.6

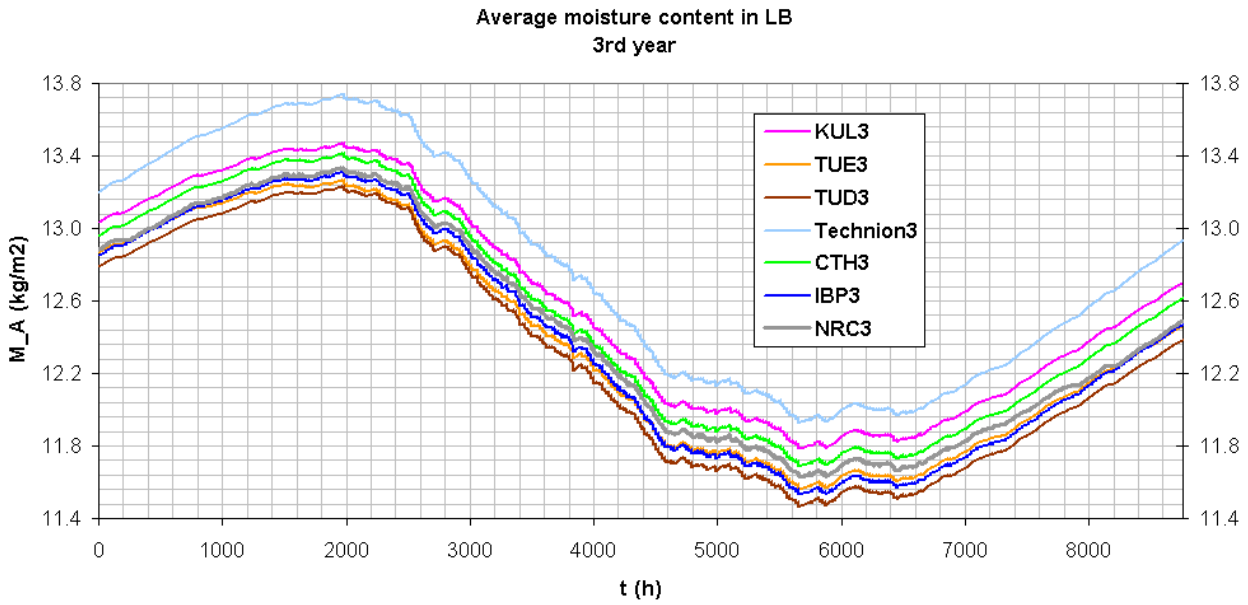


Figure 3.1.7

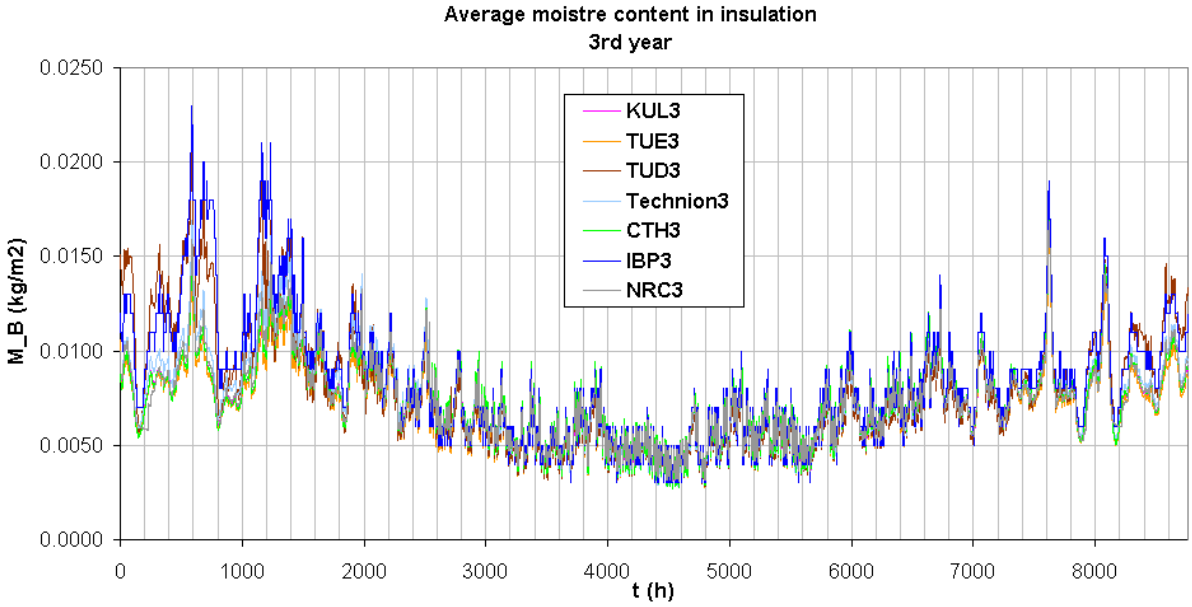


Figure 3.1.8

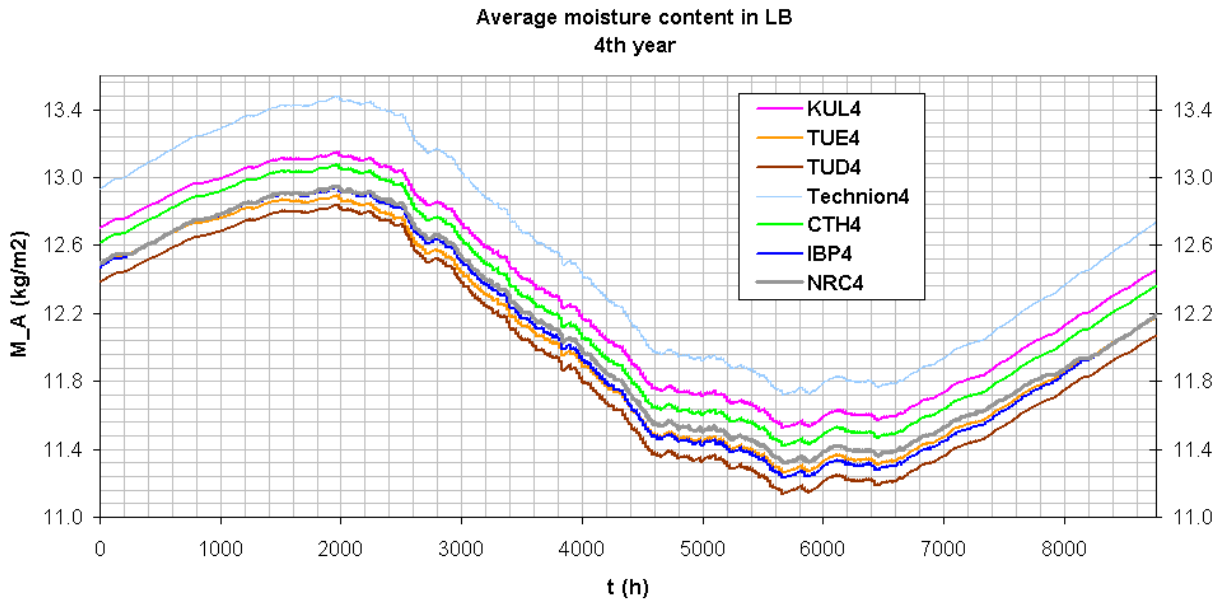


Figure 3.1.9

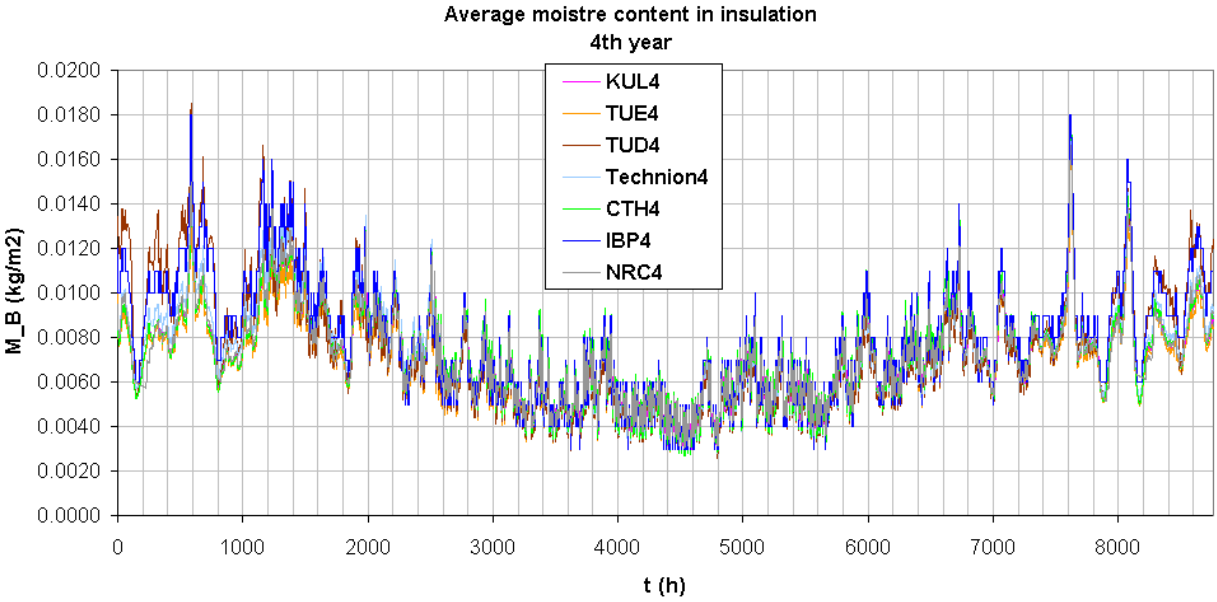


Figure 3.1.10

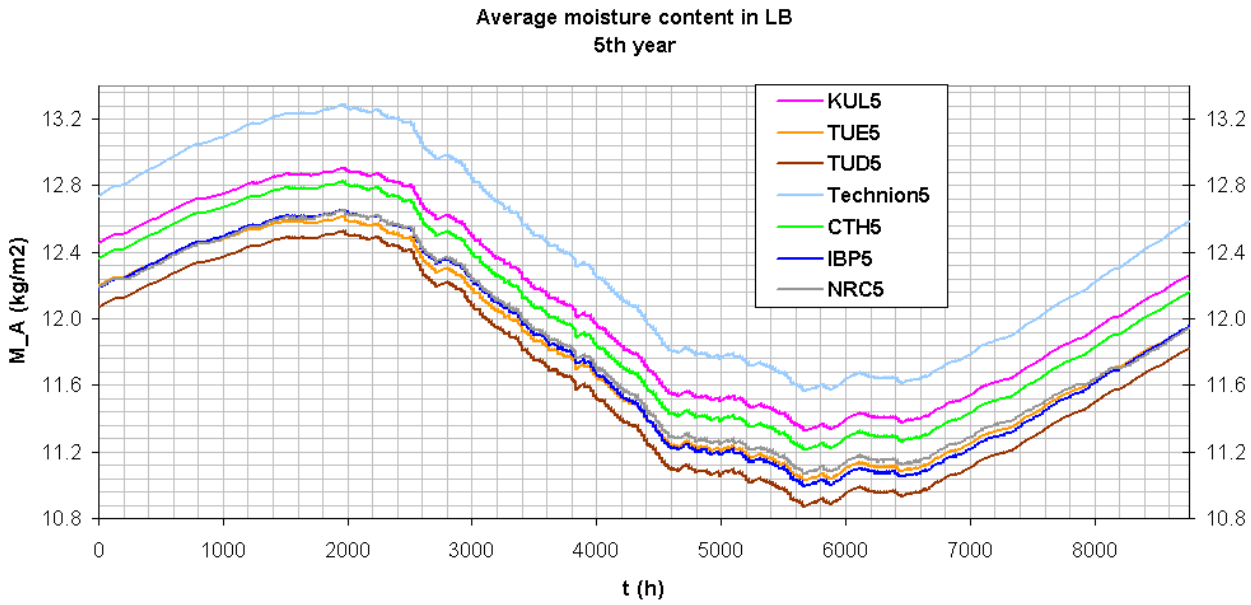


Figure 3.1.9

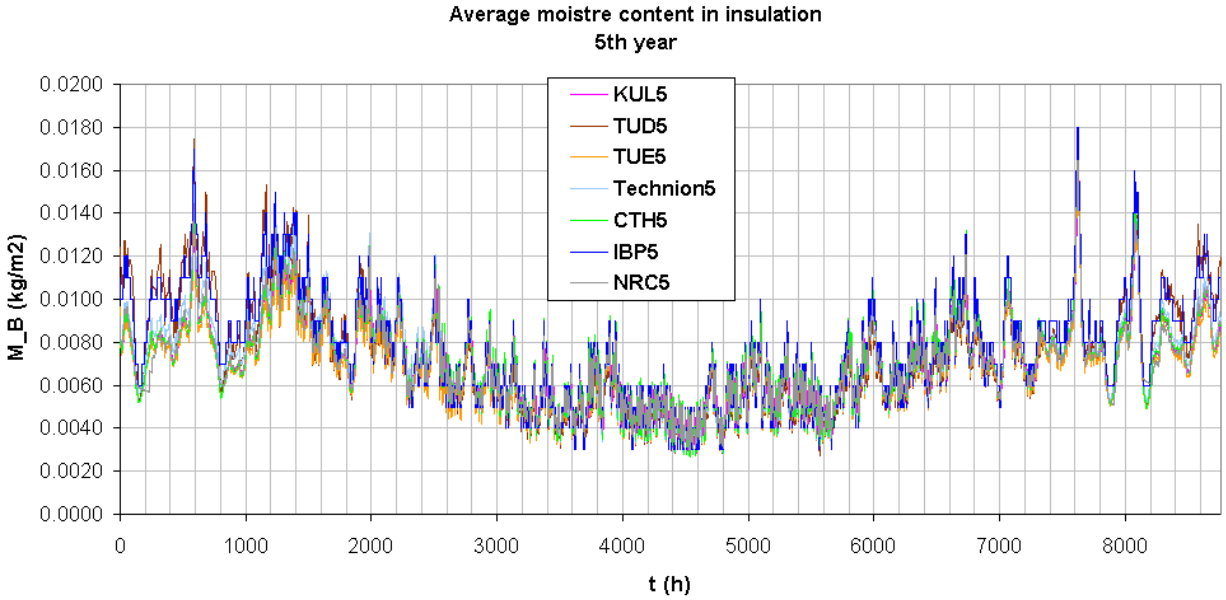


Figure 3.1.10

3.2 Benchmark two: 'Analytical Benchmark'

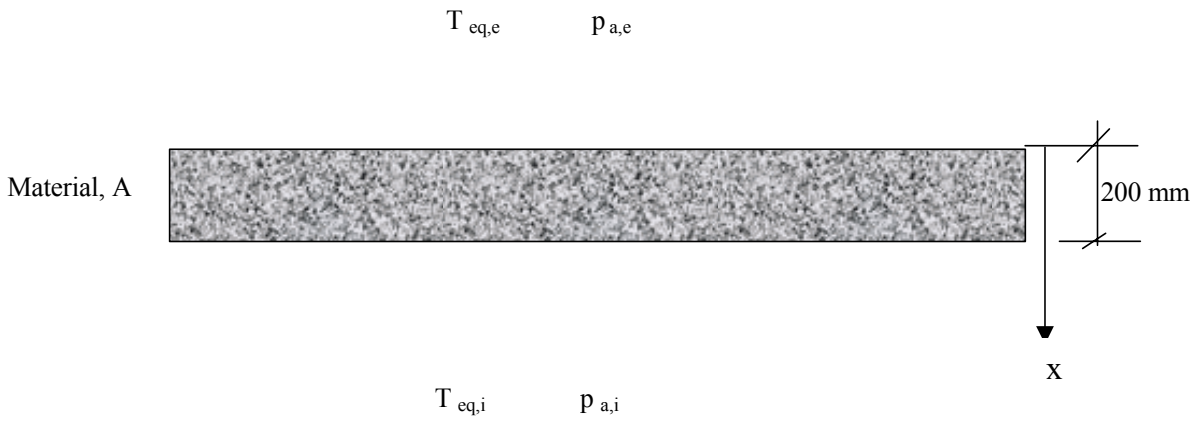
Homogeneous wall

HAMSTAD: Benchmark no: 2, Case with analytical solution

Initiated by Israel Institute of Technology (Technion)

A homogeneous layer is analyzed under isothermal conditions in one dimension. The layer is initially in moisture balance with the ambient air, having constant relative humidity. At time zero there is a sudden change in the relative humidity of the surrounding air. The structure is perfectly airtight.

Structure



Properties of layers

1 General data:

Isothermal case, decoupling of thermal and moisture process: $\ell_{\ell v}$ =56.8 J/kg

2 Material properties:

Material A

- Sorption isotherm: $w = \dfrac{116}{(1 - \dfrac{1}{0.118} \cdot \ln(\phi))^{0.869}}$ (kg/m^3)

$$\phi = \exp(0.118 \cdot (1 - (\frac{116}{w})^{\frac{1}{0.869}}))$$ (-)

- Vapour diffusion: $\delta_p = 1.0 \cdot 10^{-15}$ (s) (constant)

- Moisture diffusivity: $D_w = 6.0 \cdot 10^{-10}$ m^2/s (constant)

$$D_w = -K \frac{\partial P_{suc}}{\partial w} + \delta_p p_s \frac{1}{\xi}$$ see Equations (38-39), [1]

$$\frac{\partial P_{suc}}{\partial w} = -\frac{RT\rho_w}{M_w} \frac{1}{\phi\xi}$$ Using Kelvin's law

- Thermal conductivity: $\lambda = 0.15$ W/(mK) (constant)

- Specific heat capacity: c_p=800 J/kgK

- Density: ρ_0= 525 kg/m^3

3 Dimensions:

Thickness: 0.2 m

Boundary conditions

1 Thermal and humidity conditions:

Equiavalent temperatures: $T_{eq,e}$=20 ºC, $T_{eq,i}$=20 ºC

Equivalent temperatures accounts for both ambient air temperature and radiation exchange.

An instant drop of the air relative humidity to $\phi_{a,e}(t) = 0.45$ on exterior side and to $\phi_{a,i}(t) = 0.65$ on the interior side, i.e:

$$p_{a,e} = 0.45 \cdot p_s(20\ ^\circ\mathrm{C}) \quad p_{a,i} = 0.65 \cdot p_s(20\ ^\circ\mathrm{C}) \quad t > 0$$

Here, p_s, is the partial pressure at saturation.

2 Pressure conditions: no air pressure difference

3 Surface transfer coefficients:

$\alpha_{e,e}$ = 25 W/m^{2}K, $\alpha_{e,i}$ = 25 W/m^{2}K, $\beta_{p,e} = 1.0 \cdot 10^{-3}$ s/m , $\beta_{p,i} = 1.0 \cdot 10^{-3}$ s/m

Initial conditions

w= 84.7687 kg/m^3 (RH =95%), T=20 ºC

Output

1 Simulation time: 1 000 hours.

2 Requested output:

Results should be displayed for the cross-sectional distribution of moisture content, $w(x)$ (kg/m^3) at 100, 300 and 1000 hours.

Output file structure: in ASCII format

Values should be tabulated for x=0 m to 0.20 cm, at 0.01 m intervals, using four decimal digits, e.g:

Column 1	Column 2	Column 3	Column 4	…	Column 22
time (h)	w(0)	w(0.01)	w(0.02)	…	w(0.2)
100	…	…	…	…	…
300	…	…	…	…	…
1000	…	…	…	…	…

3 Names of data files:

The output file should be named as: 'University/Institute'+ Bench2+'Output file no.' + .txt, e.g. the file from Chalmers (CTH): CTHBench2.txt

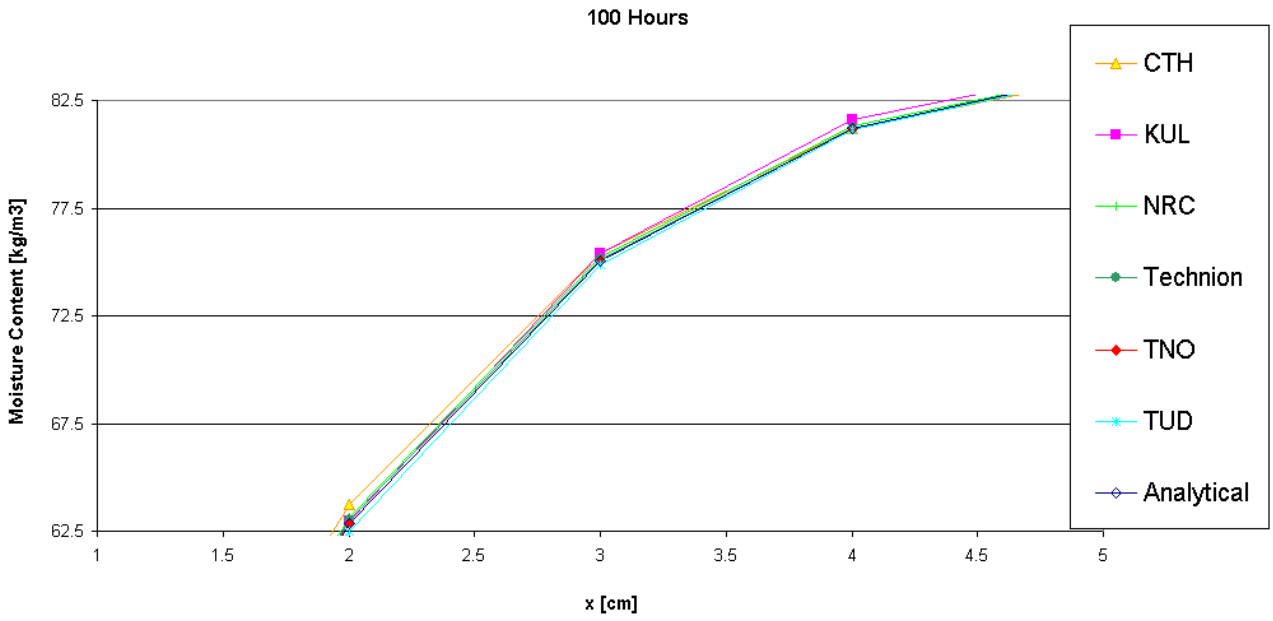


Figure 3.2.1

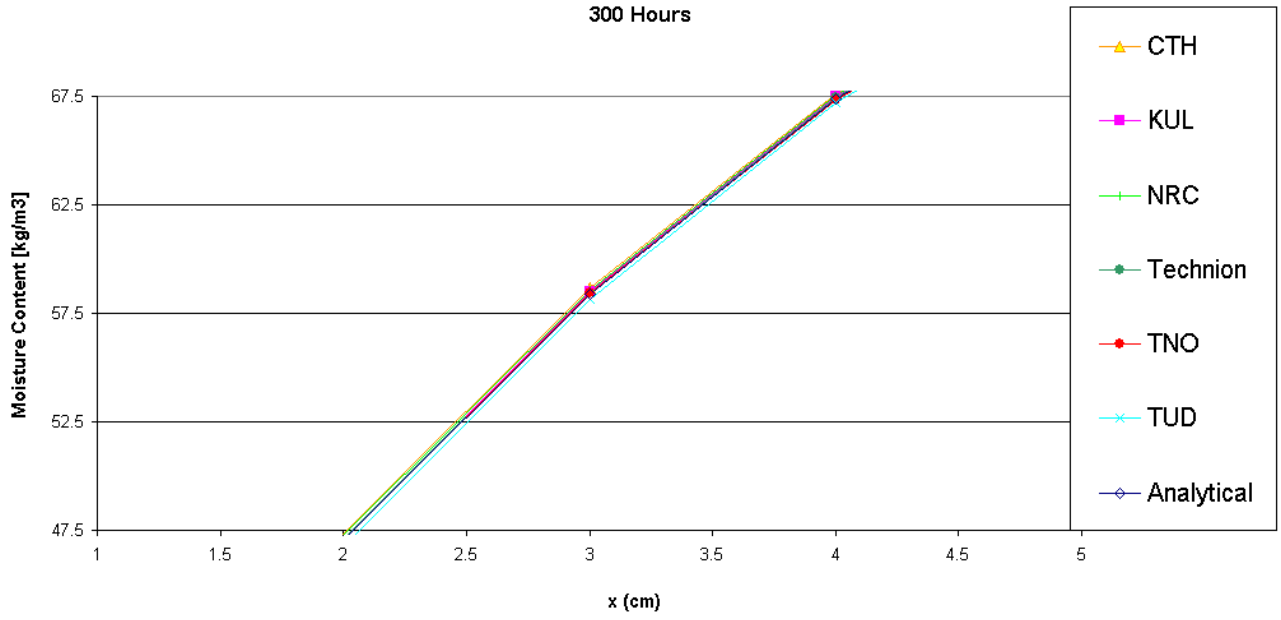


Figure 3.2.2

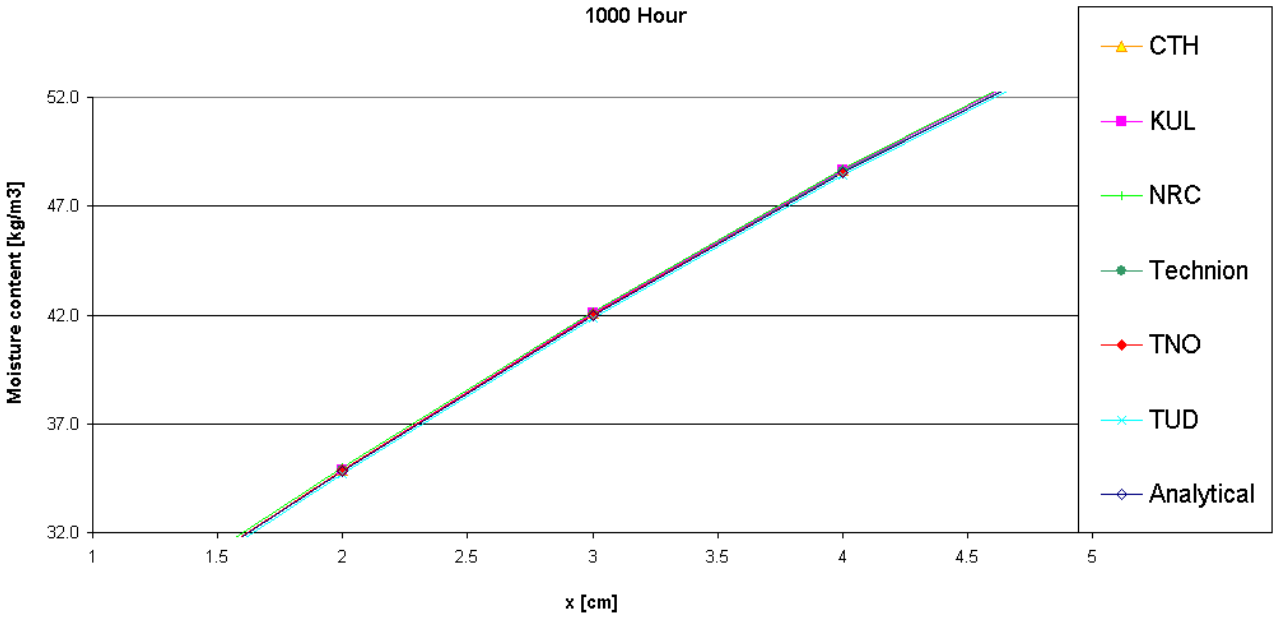


Figure 3.2.3

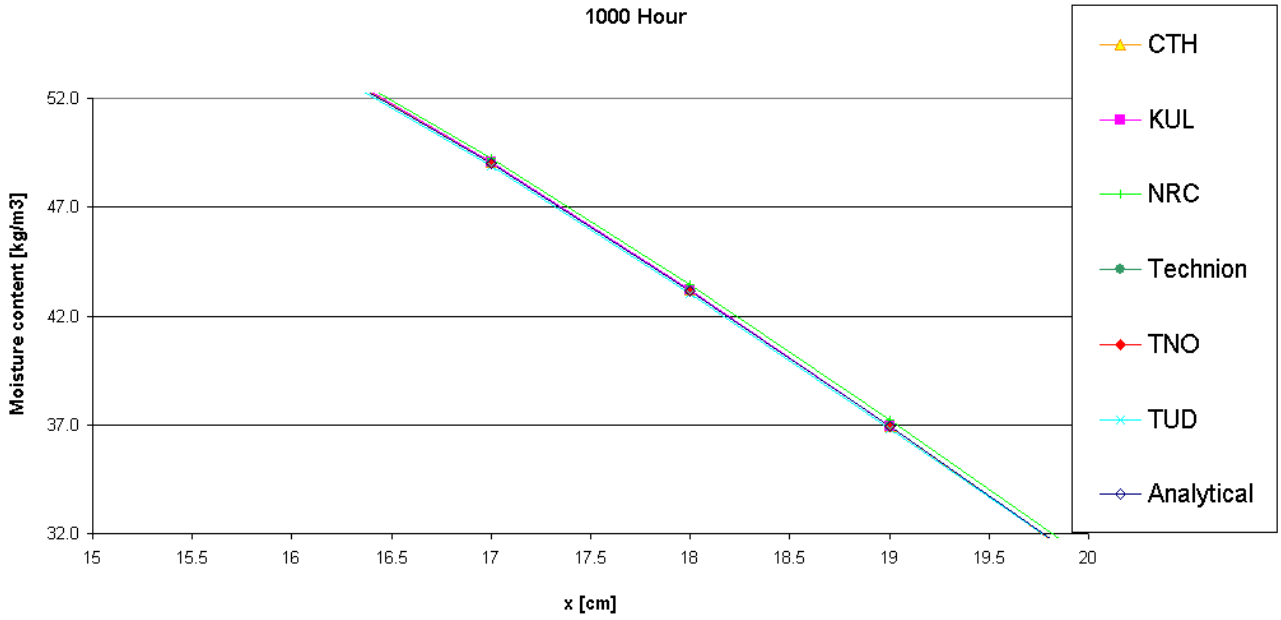


Figure 3.2.4

3.3 Benchmark three: *'Lightweight wall'*

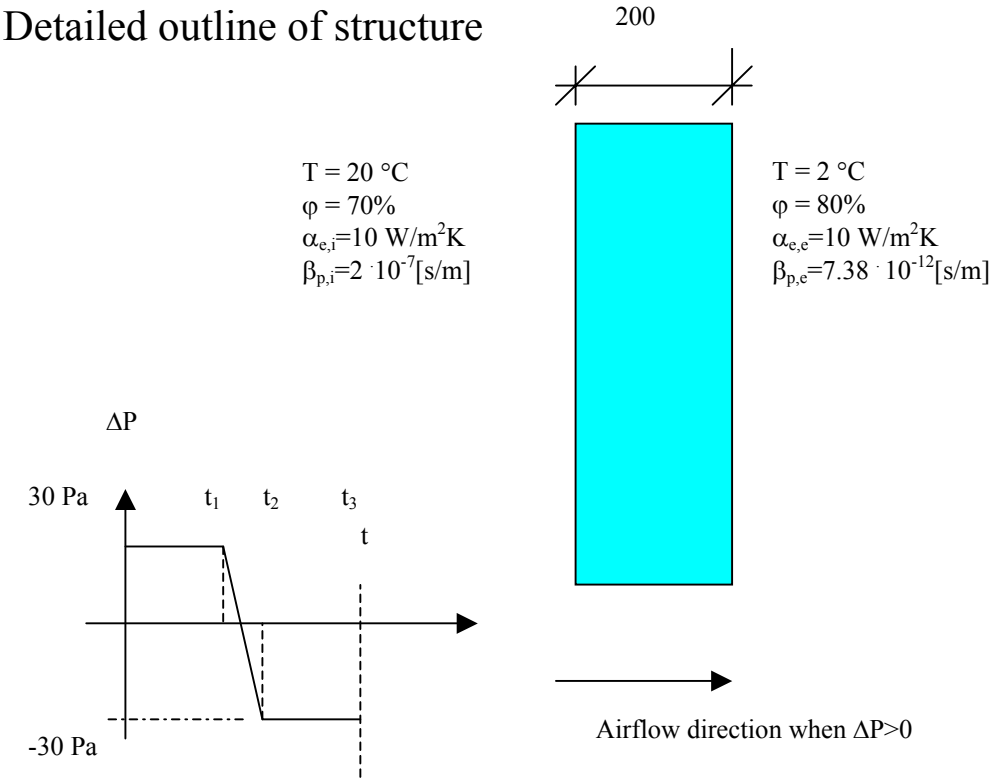
Light weight wall

HAMSTAD: Benchmark no: 3

Initiated by Chalmers University of Technology (CTH)

A single layer is analysed.

Detailed outline of structure



t_1=20 days
t_2=21 days
t_3=100 days

Input data:

1 General data / relations:

Definitions according to [1].

$T_{Ref} = 293.15$ K	Reference temperature (for parameter transformations etc.)
$\rho_w = 1000$ kg/m^3	Liquid water density
$R = 8.314$ J/(mol · K)	General gas constant
$M_w = 0.018$ kg/mol	Molar water mass
$R_v = R / M_w$	Vapour gas constant
$\ell_{\ell v} = 2.5 \cdot 10^6$ J/kg	Mass specific phase transition enthalpy water liquid < > vapour
g = 9.81 m/s²	Gravity acceleration
$h(P_{Suc}) \equiv P_{Suc} / (\rho_w \cdot g)$	Capillary suction height (m)
$P_{Suc}(\phi) = - \rho_w \cdot R_v \cdot T \cdot \ln(\phi)$	Kelvin relation
$w \equiv m_w / V_{REV}$ $\approx (m_w + m_v) / V_{REV}$	Gravimetric liquid water content (kg/m³) ≈ Total moisture (liquid + vapour) content
$\theta_l(w) = w / \rho_w$	Volumetric liquid water content (m³/m³)

2 Material property definitions:

Transport coefficient definitions according to HAMSTAD methodology.

- Water retention curve / Sorption isotherm:

$$w(P_{suc}) = w_{sat} \cdot \sum_{i=1}^{N} \frac{k_i}{(1+(a_i \cdot h(P_{suc}))^{n_i})^{m_i}} \ \text{(kg/m}^3\text{)} \ ,$$

$$m_i = 1 - 1/n_i$$

- Vapour diffusion: $\delta_p(w,T) = \frac{M_w}{RT} \cdot D_{vapour}(w)$;

$$D_{air,Schirmer}\left[\frac{m^2}{s}\right] = 2{,}31 \cdot 10^{-5} \cdot \left(\frac{T_{ref}[K]}{273.16}\right)^{1{,}81} = 26.1 \cdot 10^{-6}$$

$$D_{vapour}(w) = \frac{26.1 \cdot 10^{-6}}{\mu_{dry}} \cdot \frac{1 - \frac{w}{w_{sat}}}{(1-p) \cdot (1 - \frac{w}{w_{sat}})^2 + p} \ \text{(m}^2\text{/s)}$$

- Liquid water conductivity:

$$K(w) = \exp[\sum_{i=0}^{5} a_i \cdot (w/\rho_w)^i] \quad \text{(s)}$$

- Thermal conductivity:

$$\lambda = \left(\lambda_{Dry} + \lambda_{Mst} \frac{w}{\rho_w} \right) \text{ (W/mK)}$$

- Heat capacity for dry material: $\rho_0 c_0$ [J/m^{3}K]

Parameters for material property of single layer

Water Retention	
w_{sat}[kg/m³]	871
k_1 [--]	0.41
k_2 [--]	0.59
a_1 [1/m]	0.006
a_2 [1/m]	0.012
n_1 [--]	2.4883
n_2 [--]	2.3898
Vapour diffusion	
μ_{dry} [--]	5.6
p [--]	0.20
Liquid WatCond	
a_0 [--]	-46.245
a_1 [--]	294.506
a_2 [--]	-1439
a_3 [--]	3249
a_4 [--]	-3370
a_5 [--]	1305
Thermal Conduction	
λ_{dry} [--]	0.06
λ_{mst} [--]	0.56
Heat capacity	
ρ_0 [kg/m³]	212
c_0 [J/kgK]	1000

Air

- Heat capacity: $\rho_a c_p = 1.25$ (kJ/m^{3}K)

Note: Variation of air properties due to temperature changes are neglected.

3 Material thickness: 200 mm

4 Initial condition

T=20 °C
RH=95 %
$r_a = C \,.\, \Delta P^{\gamma}$ $C = 3.10^{-5}$ [m^3/m^2·s·Pa] $\gamma = 1$

5 Boundary conditions:

1. Thermal condition: see 'Detailed outline of structure'
2. Humidity conditions: see 'Detailed outline of structure'
3. Pressure conditions: see 'Detailed outline of structure'
4. Surface resistance conditions:

$\alpha_{e,i}$ =10 W/m^{2}K , $\alpha_{e,e}$ = 10 W/m^{2}K, $\beta_{p,i}$ =2 · 10^{-7} s/m, $\beta_{p,e}$ =7.38 ·10^{-12} s/m

Output:

1. Simulation time: 2400 hours

2. Requested output:

- Temperature [^{o}C]

T(x=0.05), T(x=0.10), T(x=0.15), T(x=0.17) and T(x=0.19).

- Moisture content [kg/m^3]

w(x=0.05), w(x=0.10), w(x=0.15), w(x=0.17) and w(x=0.19).

Output file structure: (in ASCII format)

output file one

Column 1	Column 2	Column 3	...	Column 6
time (daily)	w(x=0.05)	w(x=0.10)	...	w(x=0.19)
0	……..	……	…	…….
24	……..	……	…	…….
…				
2400	……..	……	…	…….

output file two

Column 1	Column 2	Column 3	...	Column 6
time (daily)	T(x=0.05)	T(x=0.10)	...	T(x=0.19)
0	……..	……	…	…….
24	……..	……	…	…….
…				
2400	……..	……	…	…….

3 Names of data files:

The output files should be named as: 'University/Institute'+ Bench3+'Output file no.' + .txt, e.g. files from Chalmers (CTH): CTHBenc31.txt and CTHBench32.txt

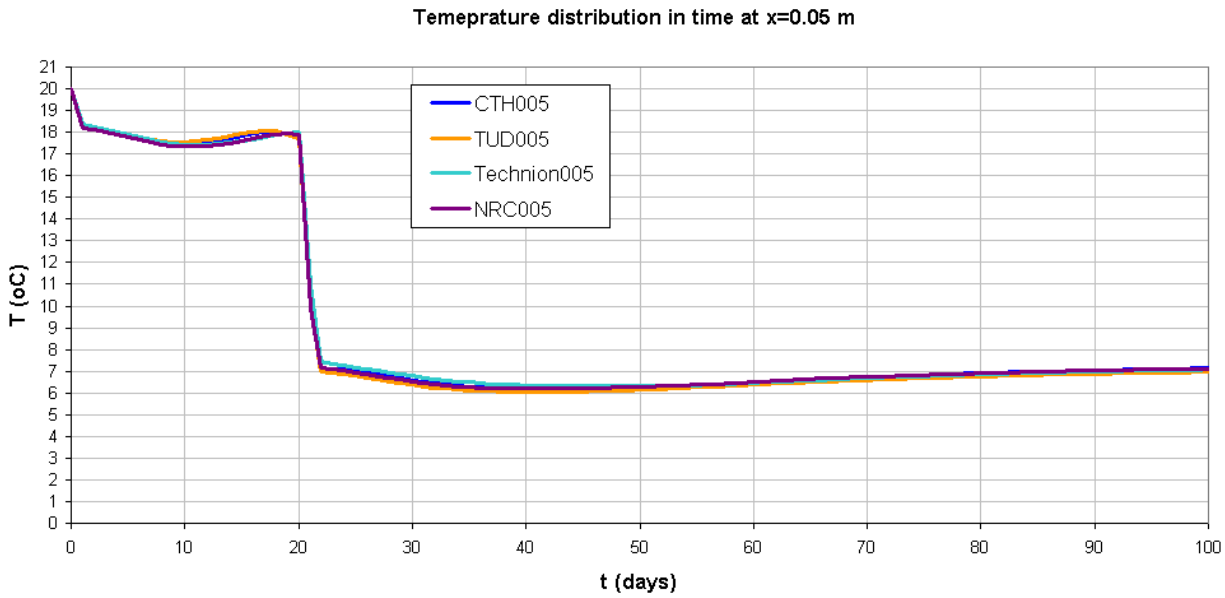


Figure 3.3.1

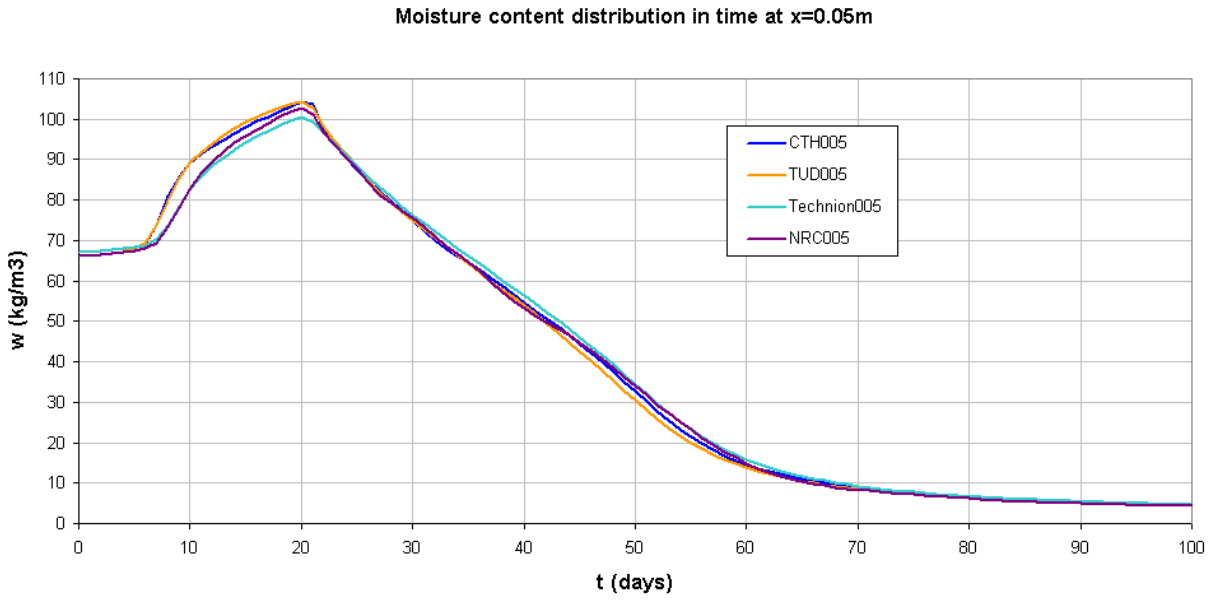


Figure 3.3.2

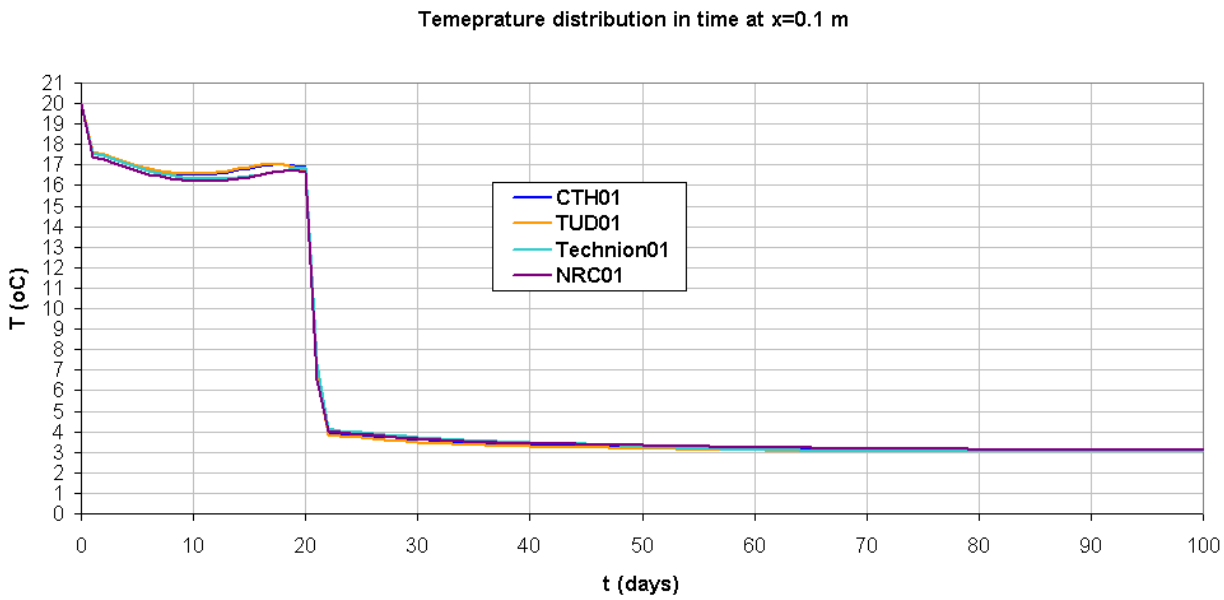


Figure 3.3.3

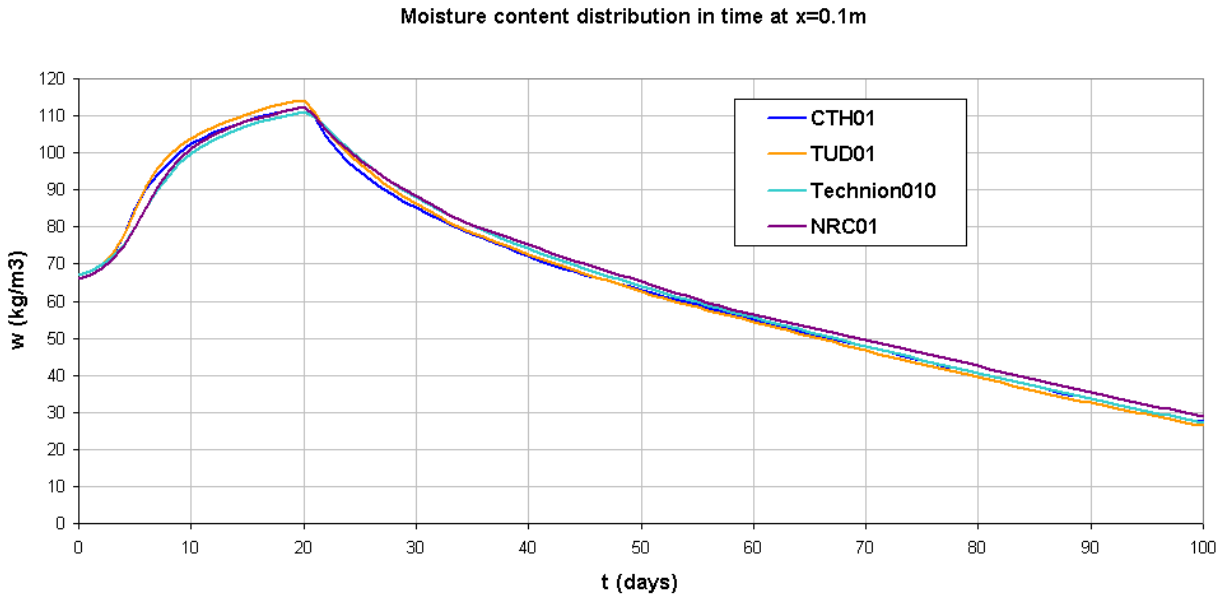


Figure 3.3.4

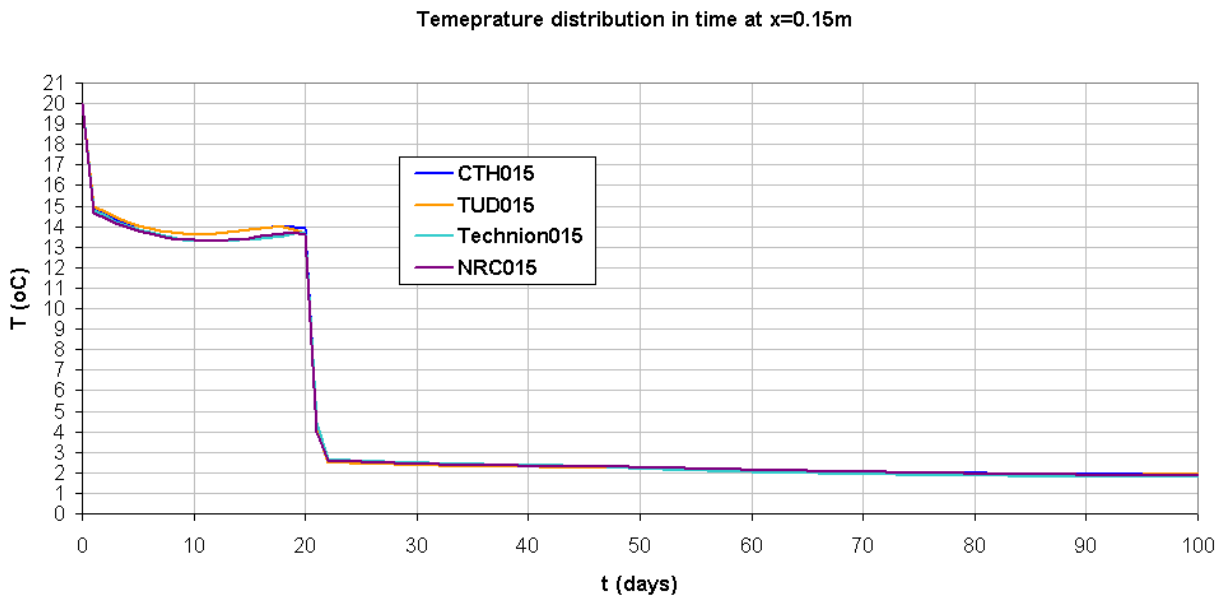


Figure 3.3.5

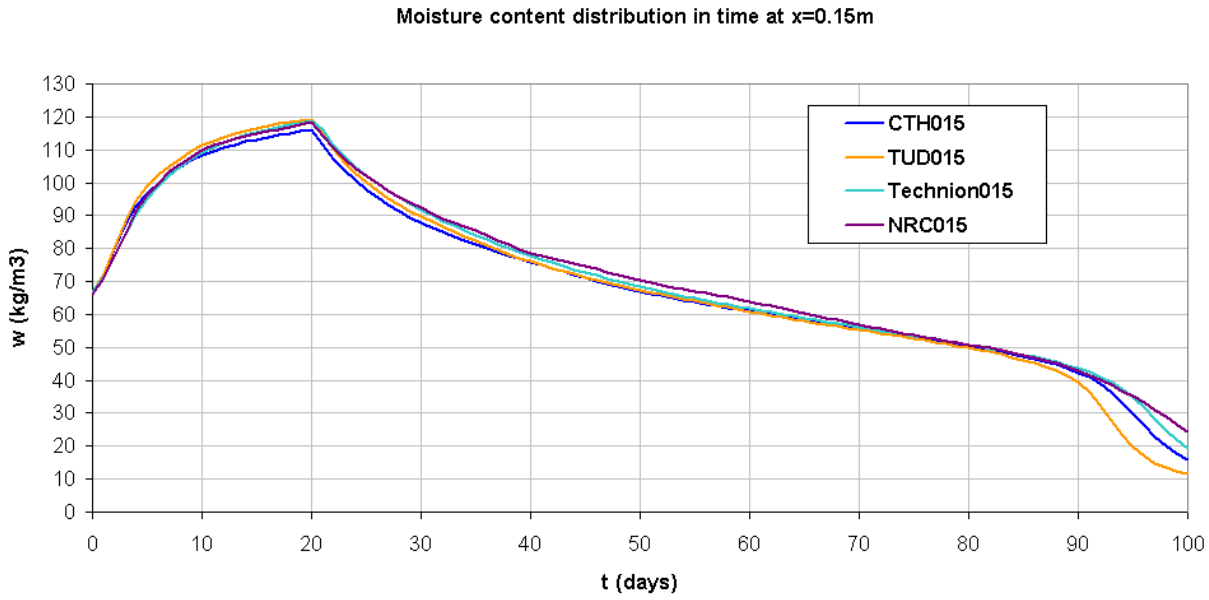


Figure 3.3.6

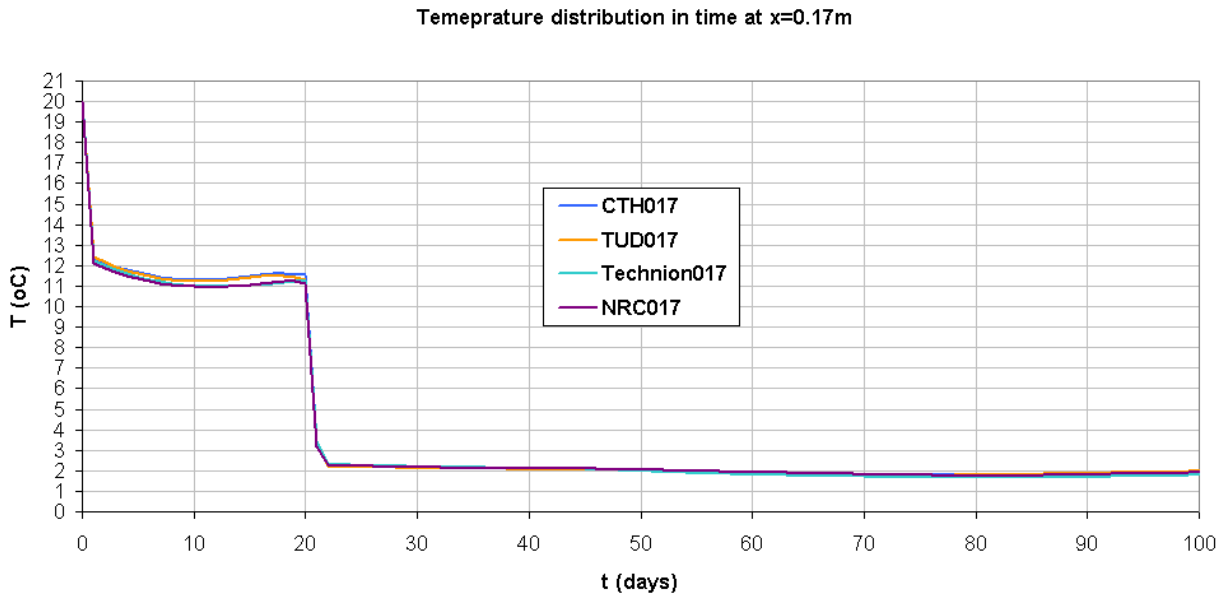


Figure 3.3.7

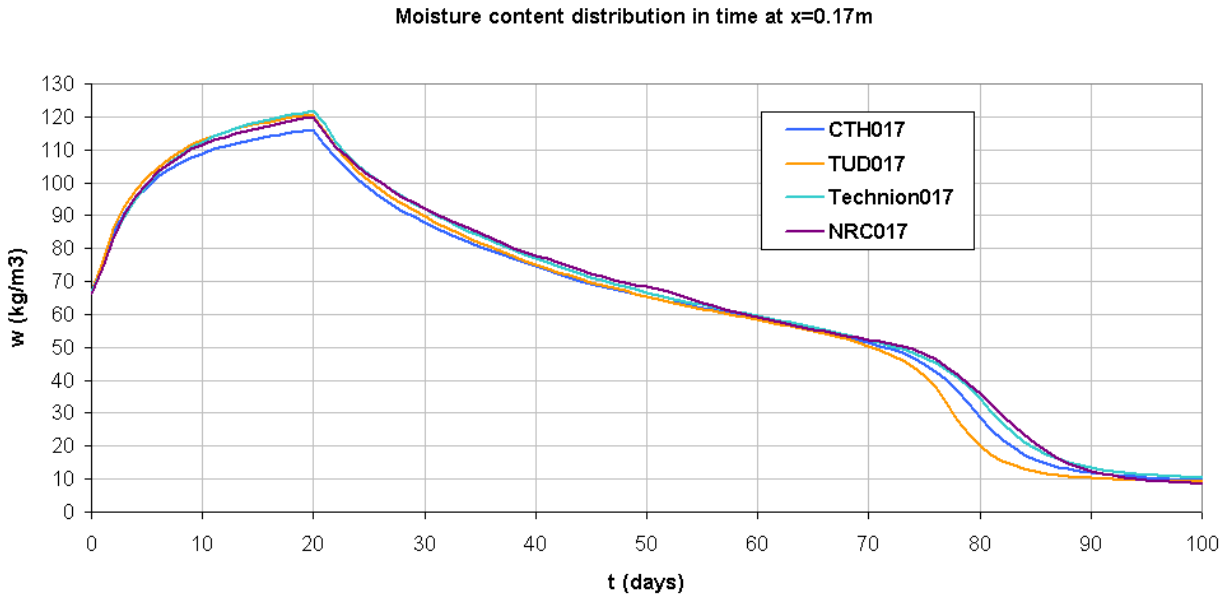


Figure 3.3.8

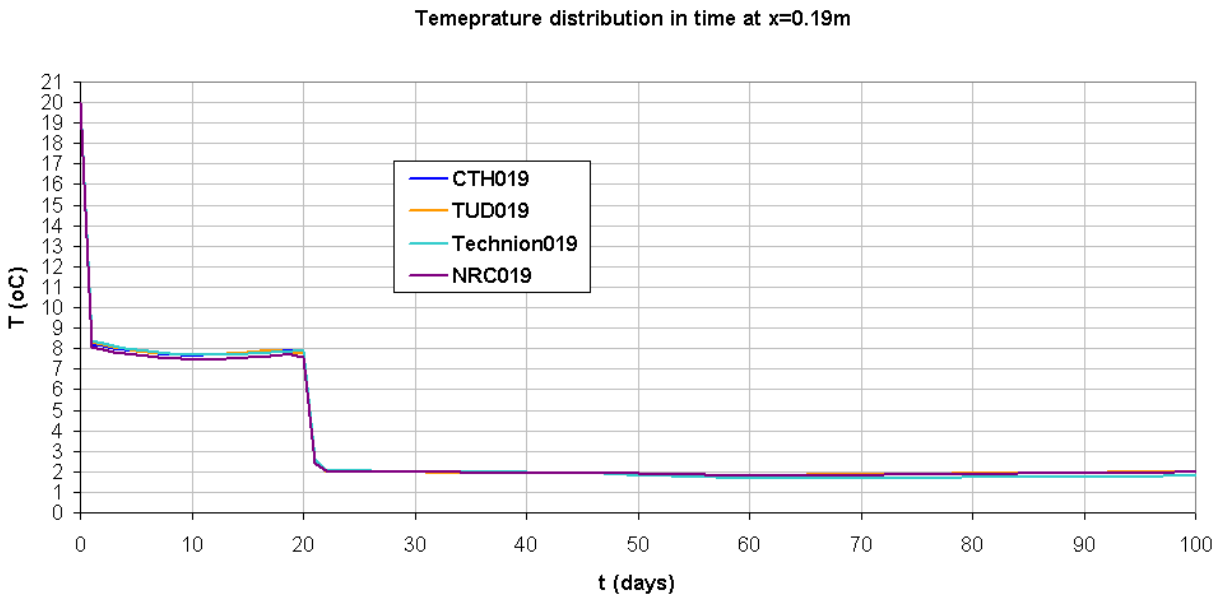


Figure 3.3.9

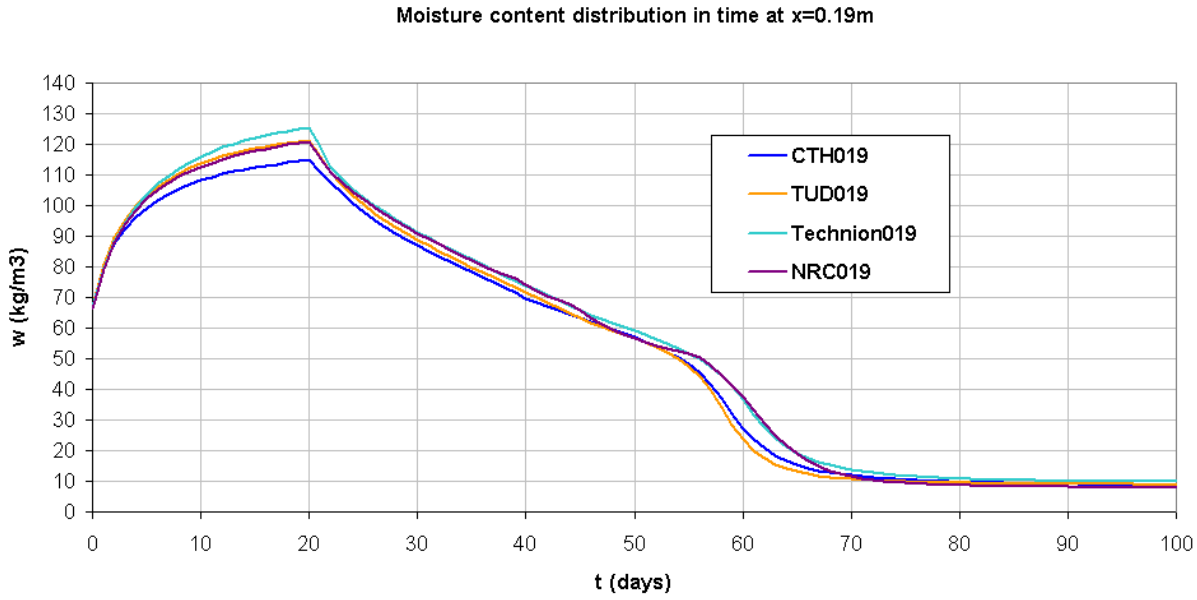


Figure 3.3.10

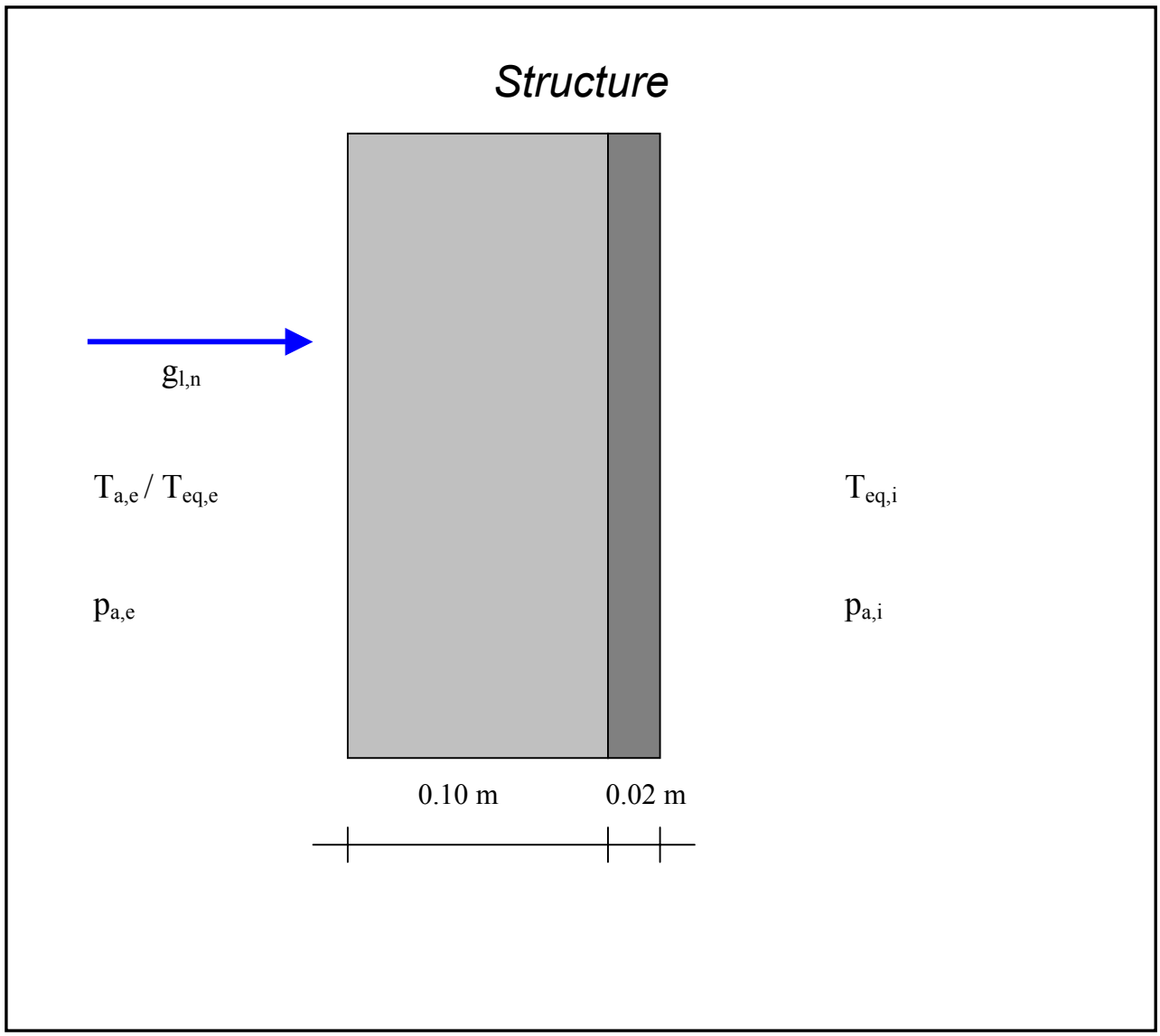
3.4 Benchmark four: *'Response analysis'*

Response analysis

HAMSTAD: Benchmark no: 4

Initiated by University of Leuven (KUL)

A wall with a hygroscopic finishing material at the inside is analysed in one dimension. The wall is submitted to subsequent relative humidity, heat and moisture loads at the inner and outer surface. The structure is perfectly airtight.



Properties of layers

1. General data:

$T = T_{ref} = 293.15$ K, $\rho_w = 1000$ kg/m³, $R = 8.314$ J/(mol,K), $M_w = 0.018$ kg/mol, $l_{lv} = 2{,}5.10^6$ J/kg

2. Material properties:

A. load bearing material

- Water retention curve:

$$w(P_{suc}) = 157.(\sum_{s_i=1}^{s_n} l_i \left(1 + (c_i P_{suc})^{n_i}\right)^{\left(\frac{1-n_i}{n_i}\right)}) \quad (\text{kg/m}^3)$$

where

$s_n = 2$
$l_1 = 0.3$; $l_2 = 0.7$
$c_1 = 1.25*10^{-5}$; $c_2 = 1.80*10^{-5}$
$n_1 = 1.65$; $n_2 = 6.0$

- Vapour diffusion:

$$\delta_p = \frac{M_w}{RT} \frac{26.1*10^{-6}}{30} \frac{1 - \frac{w}{157}}{0.503*(1-\frac{w}{157})^2 + 0.497} \quad (\text{s})$$

- Liquid water permeability:

The liquid water permeability is given as a data-file (perm.txt) containing two columns: log(P_{suc}) en log(K). Intermediate values have to be calculated via logarithmic interpolation.

- Thermal conductivity:

$$\lambda = 0.5 + 0.0045 * w \quad (\text{W/mK})$$

- Heat capacity for dry material:

$$\rho_0 c_0 = 2005 * 840 \quad (\text{J/m}^3\text{K})$$

B. finishing material at inside

- Water retention curve:

$$w(P_{suc}) = 209.(\sum_{s_i=1}^{s_n} l_i \left(1 + \left(c_i P_{suc}\right)^{n_i}\right)^{\left(\frac{1-n_i}{n_i}\right)}) \quad (\text{kg/m}^3)$$

where

$s_n = 1$
$l_1 = 1.0$
$c_1 = 2.0*10^{-6}$
$n_1 = 1.27$

- Vapour diffusion:

$$\delta_p = \frac{M_w}{RT} \frac{26.1*10^{-6}}{3} \frac{1 - \frac{w}{209}}{0.503*(1 - \frac{w}{209})^2 + 0.497} \qquad (\text{s})$$

- Liquid water permeability:

$$K = \exp(-33.0 + 0.0704*(w-120) - 1.742*10^{-4}*(w-120)^2 - 2.7953*10^{-6}*(w-120)^3 - 1.1566*10^{-7}*(w-120)^4 + 2.5969*10^{-9}*(w-120)^5) \quad (\text{s})$$

- Thermal conductivity:

$$\lambda = 0.2 + 0.0045 * w \qquad (\text{W/mK})$$

- Heat capacity for dry material:

$$\rho_0 c_0 = 790 * 870 \qquad (\text{J/m}^3\text{K})$$

3. Dimensions:

A Dimensions of the load bearing material: 0.10 m
B Dimensions of the finishing material at inside: 0.02 m

Boundary and initial conditions

1. Thermal and humidity conditions:

According to the attached data file: *ClimateBench4.txt*. Hourly values covering 5 days. Equivalent air temperatures account for both ambient air temperature and short and long wave radiation exchange. The ambient air temperatures of the outside are given to calculate the heat flux to the surface due to liquid moisture flowing into the structure (equation 49, [1]).

The climatic data-file is structured as:

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Time (s)	$T_{a,e}$ (°C)	$T_{eq,e}$ (°C)	$T_{eq,i}$ (°C)	$P_{a,e}$ (Pa)	$P_{a,i}$ (Pa)	g_l (kg/m²s)

It has to be read as follows: the first line corresponds to the climatic data between hour 0 and hour 1, second line between hour 1 and hour 2, etc... The climate is assumed to be constant during the hour. Put otherwise: climatic data is only changing stepwise at hourly values. This means that no (linear) interpolation has to be performed between the subsequent hours for each in between time step.

2. **Pressure conditions:** no air pressure difference.

3. **Surface transfer coefficients:**

   Constant transfer coefficients at interior and exterior side:

   $\alpha_{a,e} = 25$ W/m²K $\beta_{p,e} = 2*10^{-7}$ s/m

   $\alpha_{a,i} = 8$ W/m²K $\beta_{p,i} = 3*10^{-8}$ s/m

4. **Initial conditions:**

   $P_{suc} = 120738829$ Pa (w ≅ 0.4 kg/m³ for layer A)
   (w ≅ 47.5 kg/m³ for layer B)

   T=20 °C

Output

1. **Simulation time: 5 days**

2. **Requested output:**

All output files have ASCII format. Data are separated by a space.

Output file 1: (e.g. KULBench5_a.txt) gives hourly values of temperature at the outer and inner surface of the structure and of the moisture content corresponding to the capillary pressure at the outer and inner surface of the structure.

Column 1	Column 2	Column 3	Column 4	Column 5
Time(h)	w(0.00)	w(0.12)	t(0.00)	t(0.12)
0	0.4	47.5	20	20
1	…	…	…	…
…	…			
120	…			

Output file 2-21: (e.g. KULBench5_b1.txt until KULBench5_b20.txt) give the moisture and temperature profiles inside the structure every 6 hours. One file for each 6-hour-step! The co-ordinates correspond to the mesh used in the simulations.

Column 1	Column 2	Column 3
Co-ordinate (m)	w (kg/m³)	t (°C)
0.00	0.4	20
0.01	…	…
0.02	…	…
…	…	…
0.12		

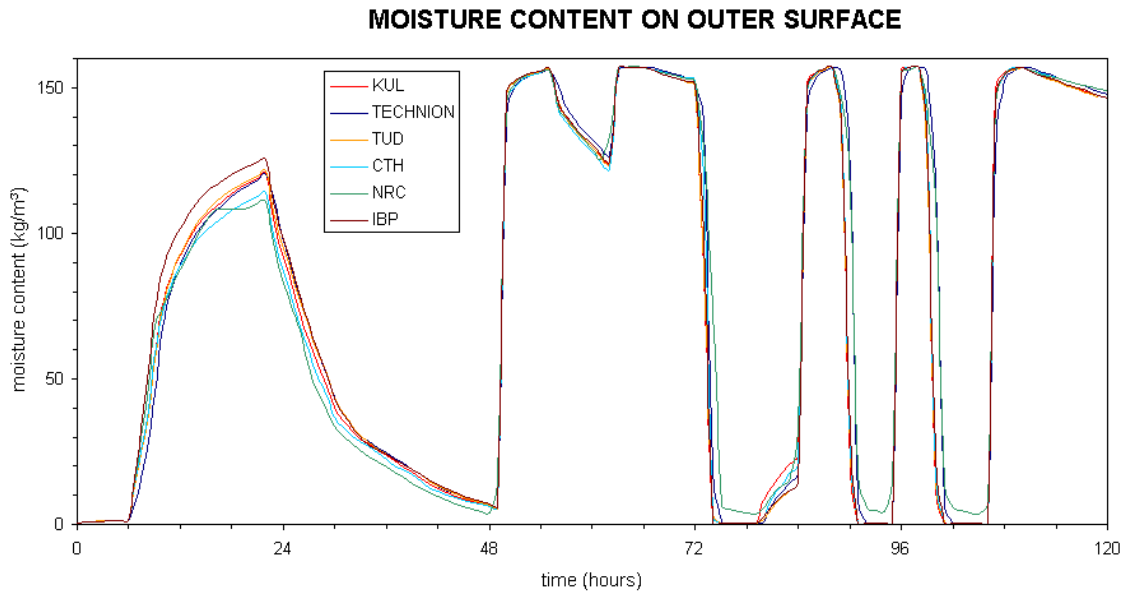


Figure 3.4.1

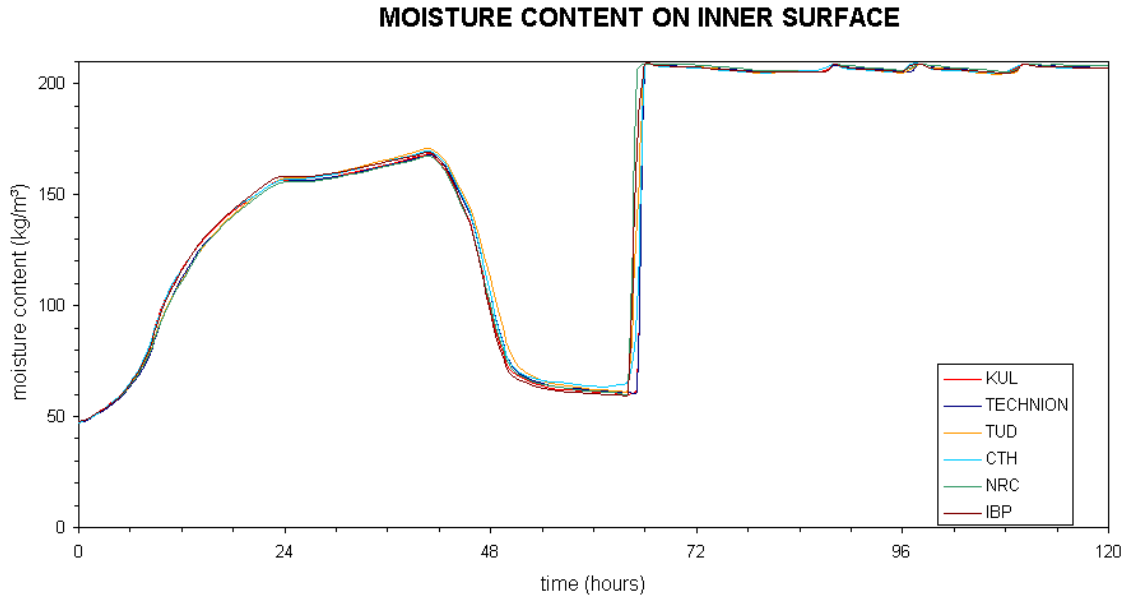


Figure 3.4.2

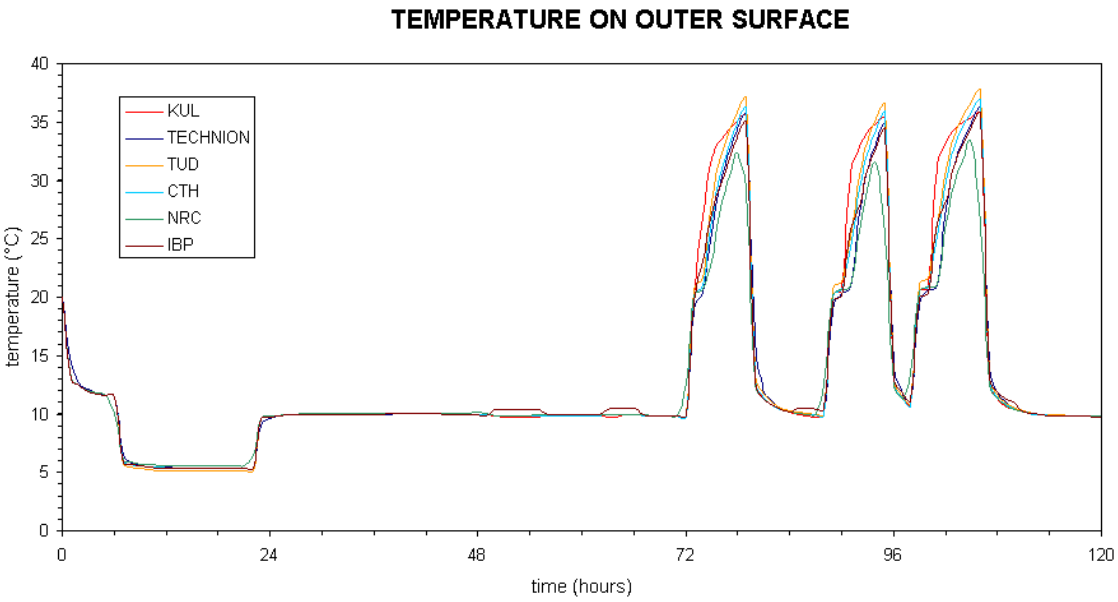


Figure 3.4.3

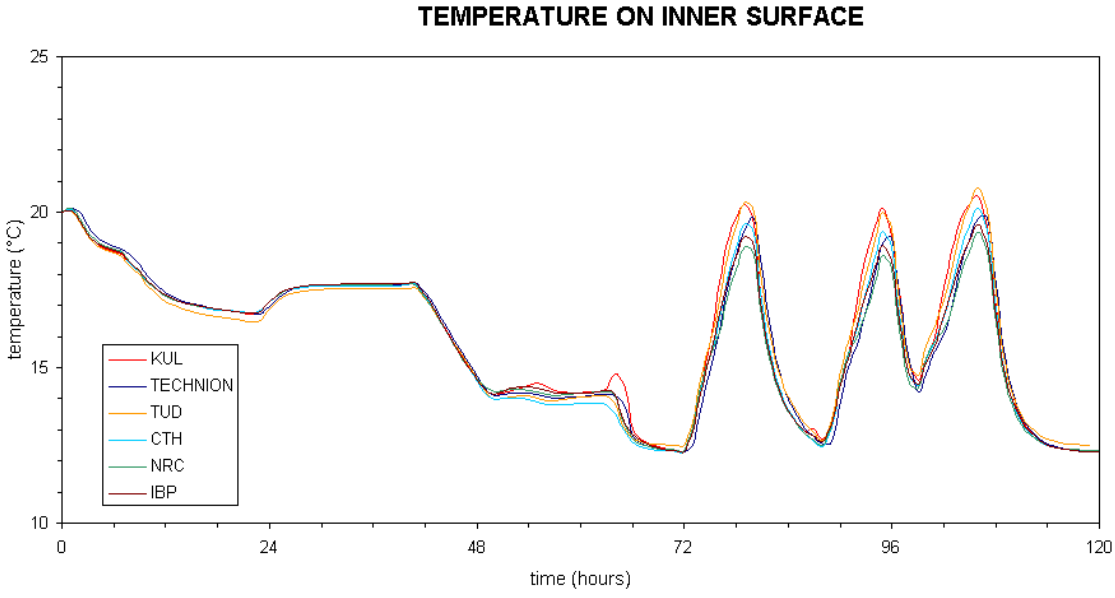


Figure 3.4.4

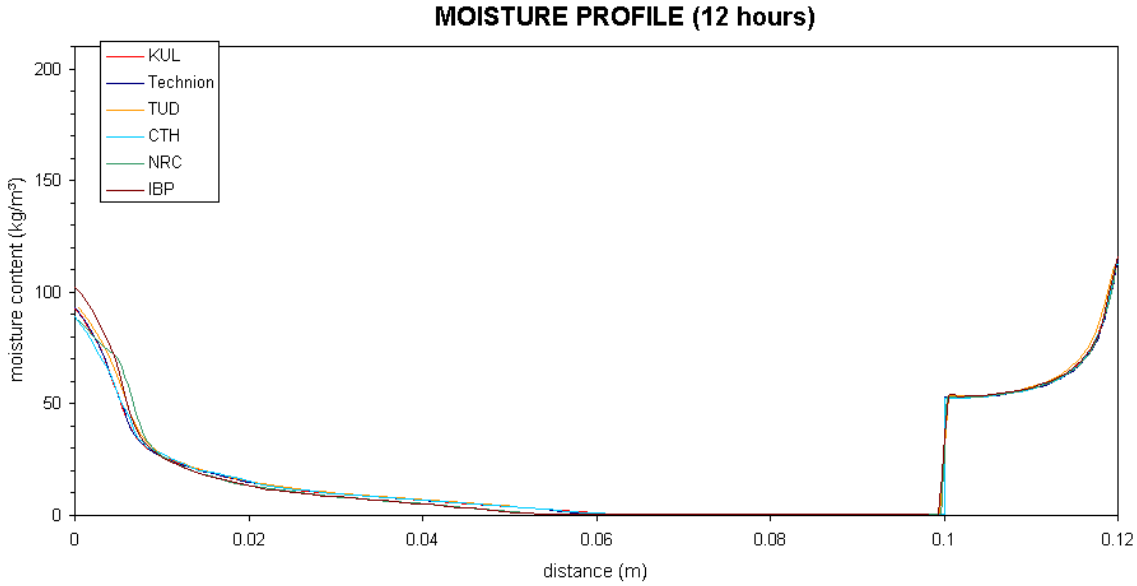


Figure 3.4.5

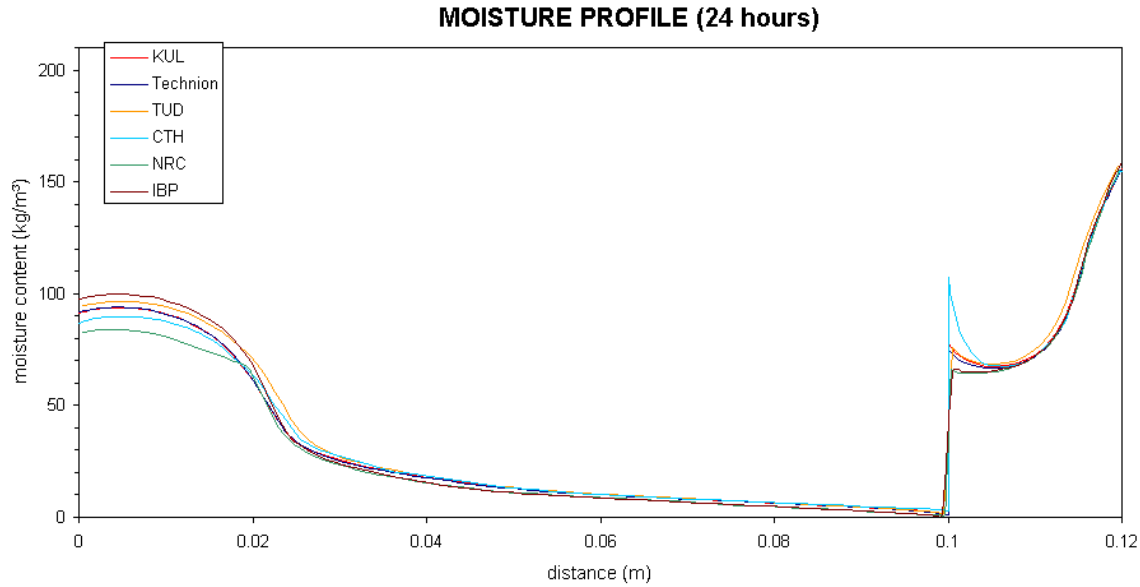


Figure 3.4.6

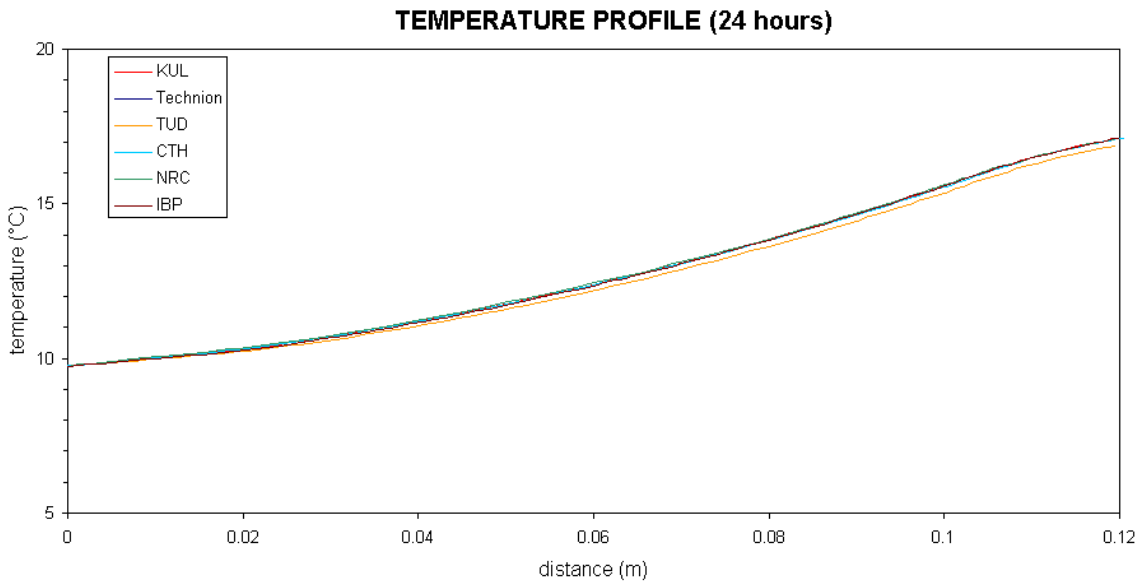


Figure 3.4.7

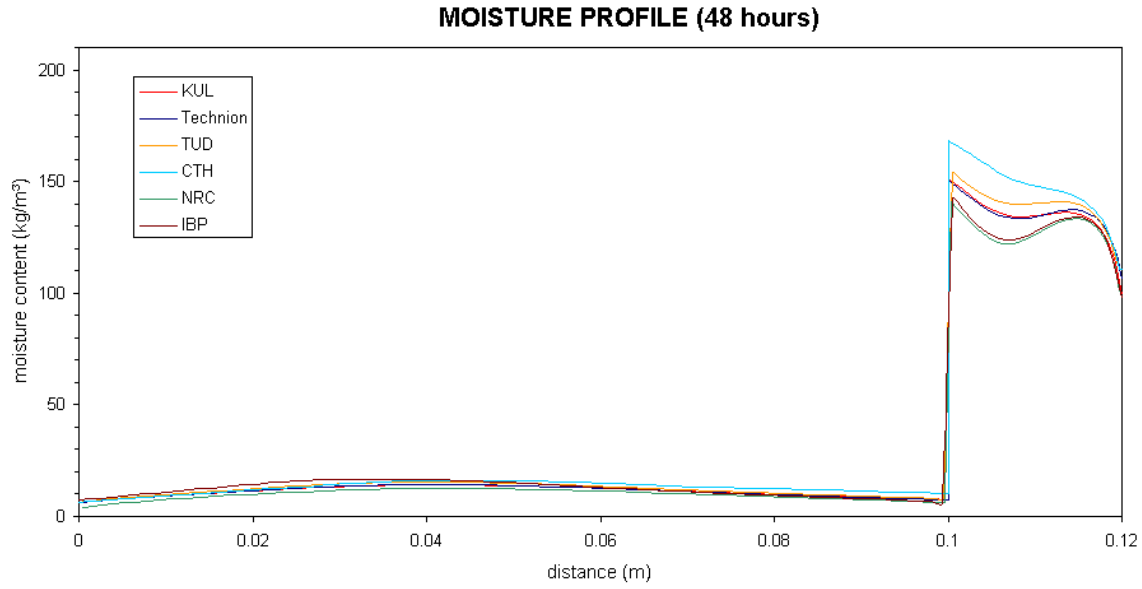


Figure 3.4.8

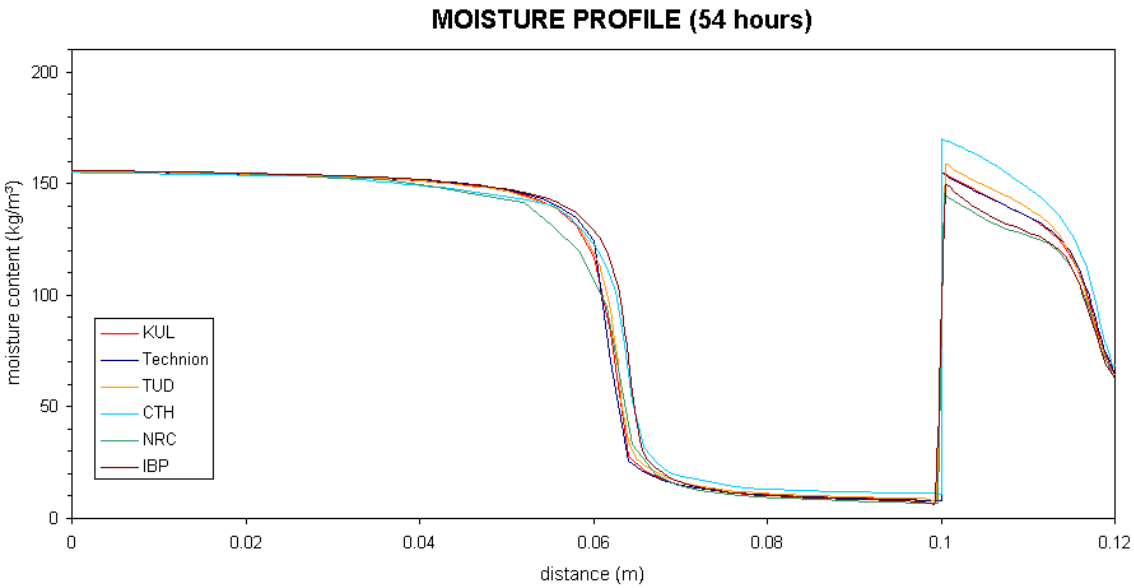


Figure 3.4.9

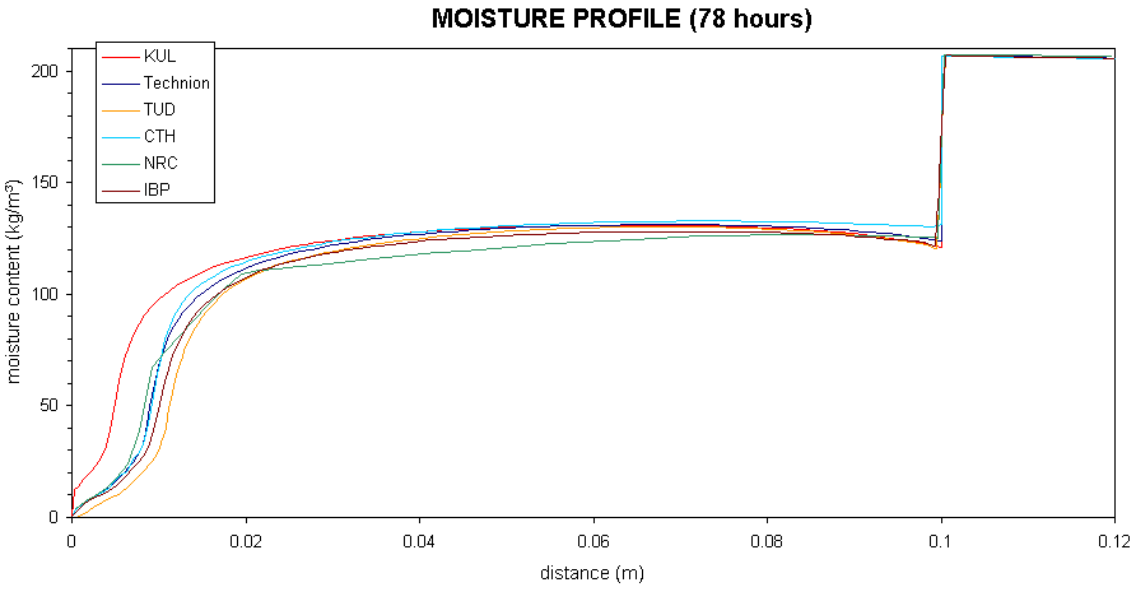


Figure 3.4.10

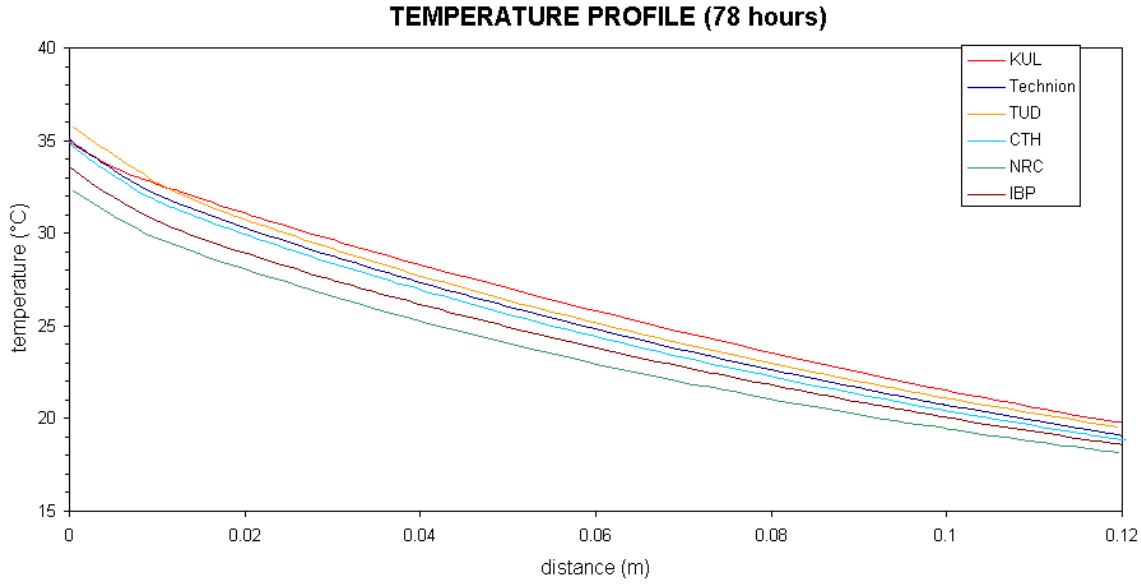


Figure 3.4.11

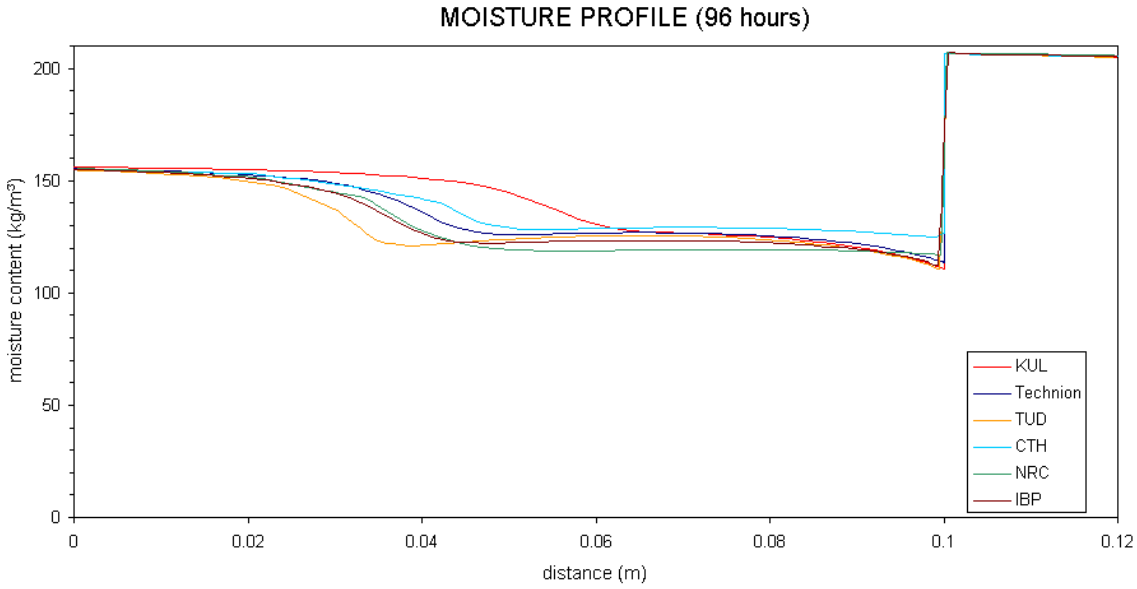


Figure 3.4.12

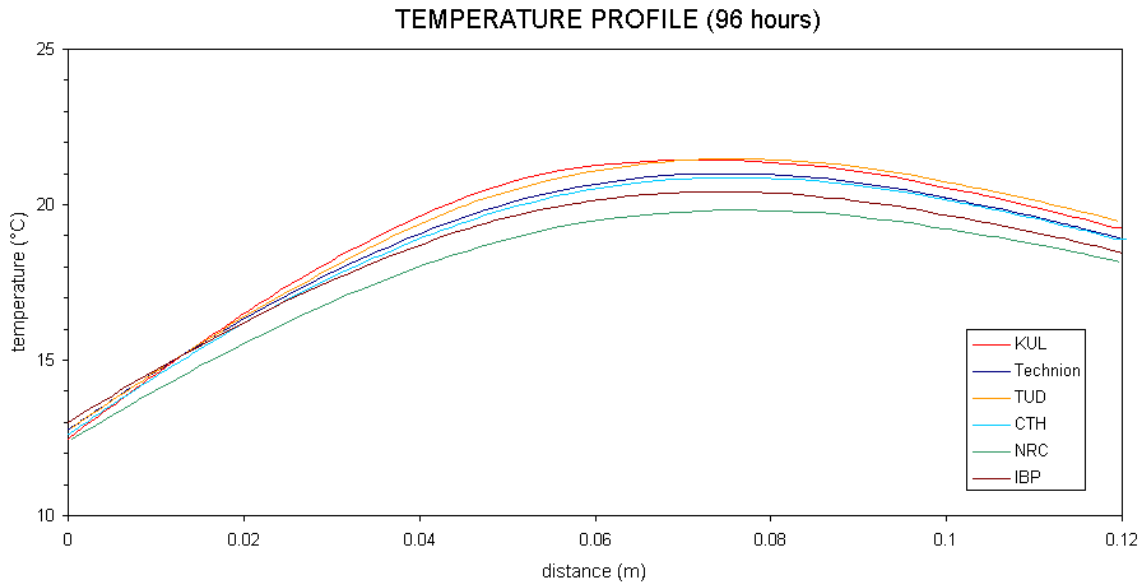


Figure 3.4.13

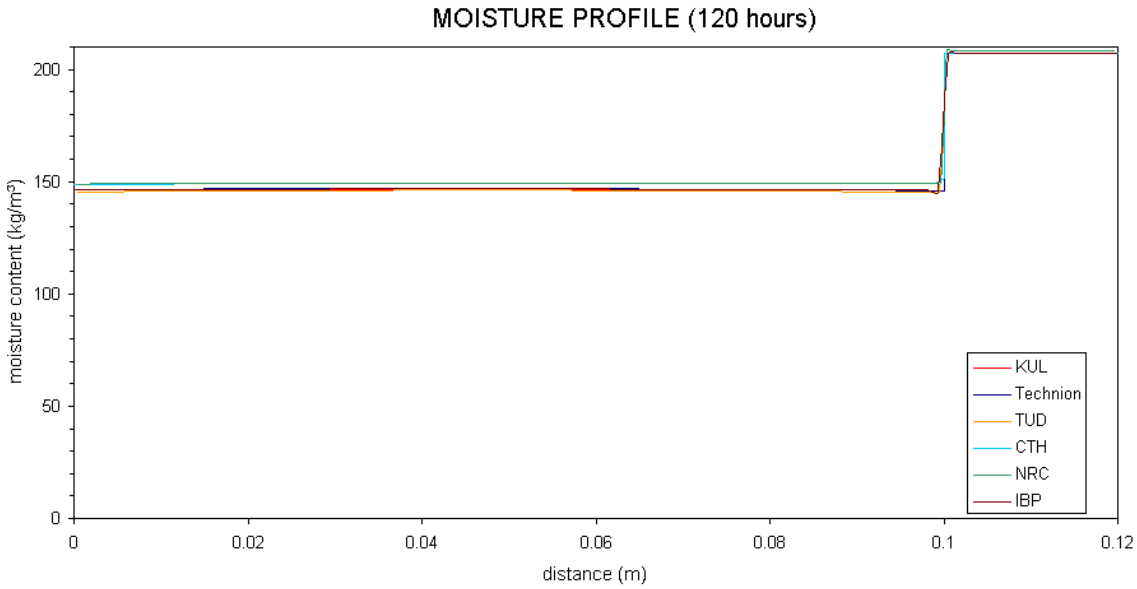


Figure 3.4.14

Benchmark five: 'Capillary active inside insulation'

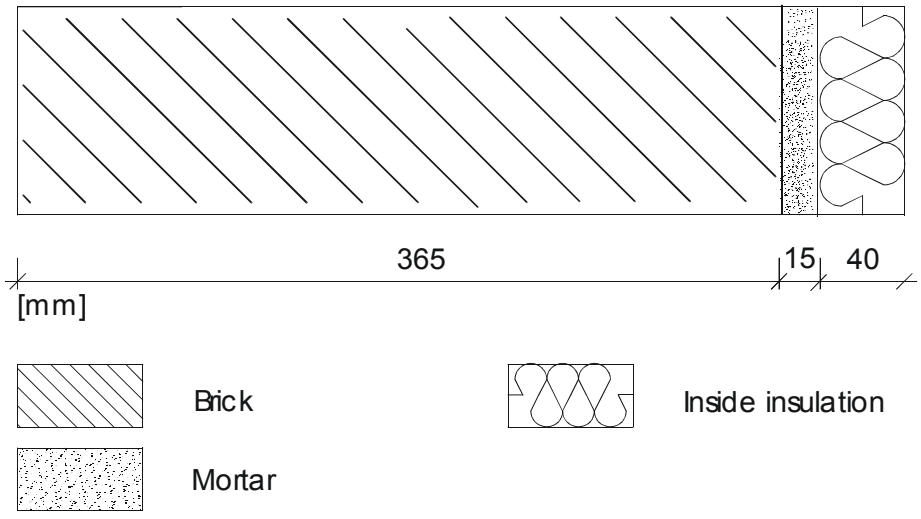
Capillary active inside insulation

HAMSTAD: Benchmark 5

Initiated by Technical University of Dresden (TUD)

An inside insulated wall structure is analysed in one dimension.
Properties of capillary active insulations are calculated.

Structure



Properties of layers

1 General data / relations:

Definitions according to [1].

$T_{Ref} = 293.15$ K	Reference temperature (for parameter transformations etc.)
$\rho_w = 1000$ kg/m^3	Liquid water density
$R = 8.314$ J/(mol · K)	General gas constant
$M_w = 0.018$ kg/mol	Molar water mass
$R_v = R / M_w$	Vapour gas constant
$\ell_{lv} = 2.5 \cdot 10^6$ J/kg	Mass specific phase transition enthalpy water liquid < > vapour
$g = 9.81$ m/s²	Gravity acceleration

$h(P_{Suc}) \equiv P_{Suc} / (\rho_w \cdot g)$ Capillary suction height (m)

$P_{Suc}(\phi) = - \rho_w \cdot R_v \cdot T \cdot \ln(\phi)$ Kelvin relation

$w \equiv m_w / V_{REV}$ Gravimetric liquid water content (kg/m³)

$\theta_l(w) = w / \rho_w$ Volumetric liquid water content (m³/m³)

2 Material property definitions:

Transport coefficient definitions according to [1].

- Water retention curve / sorption isotherm:

$$w(P_{suc}) = w_{sat} \cdot \sum_{i=1}^{N} \frac{k_i}{(1+(a_i \cdot h(P_{suc}))^{n_i})^{m_i}} \quad (\text{kg/m}^3)$$

$$m_i = 1 - 1/n_i$$

Conversion between retention and sorption via $w(P_{suc}(\phi))$ or $w(\phi(P_{suc}))$ by using Kelvin relation (see general relations above)

- Vapour diffusion: $\delta_p(w,T) = \frac{M_w}{RT} \cdot D_{vapour}(w)$;

$$D_{air,Schirmer}\left[\tfrac{m^2}{s}\right] = 2{,}31 \cdot 10^{-5} \cdot \left(\frac{T_{ref}[K]}{273.16}\right)^{1{,}81} = 26.1 \cdot 10^{-6}$$

$$D_{vapour}(w) = \frac{26.1 \cdot 10^{-6}}{\mu_{dry}} \cdot \frac{1 - \frac{w}{w_{sat}}}{(1-p) \cdot (1 - \frac{w}{w_{sat}})^2 + p} \quad (\text{m}^2/\text{s})$$

- Liquid water conductivity:

$$K(w) = \exp[\sum_{i=0}^{5} a_i \cdot (w/\rho_w)^i] \quad \text{(s)}$$

- Thermal conductivity:

$$\lambda = \left(\lambda_{Dry} + \lambda_{Mst} \frac{w}{\rho_w} \right) \text{ (W/mK)}$$

- Heat capacity for dry material: $\rho_0 c_0$ [J/m^{3}K]

3 Parameters for all given materials:

	Brick	*Mortar*	*Inside insulation*
Water Retention			
w_{sat}[kg/m³]	373.5	700	871
k_1 [--]	0.46	0.2	0.41
k_2 [--]	0.54	0.8	0.59
a_1 [1/m]	0.47	0. 5	0.006
a_2 [1/m]	0.2	0.004	0.012
n_1 [--]	1.5	1.5	2.5
n_2 [--]	3.8	3.8	2.4
Vapour diffusion			
μ_{dry} [--]	7.5	50	5.6
p [--]	0.20	0.20	0.20
Liquid WatCond			
a_0 [--]	-36.484	-40.425	-46.245
a_1 [--]	461.325	83.319	294.506
a_2 [--]	-5240	-175.961	-1439
a_3 [--]	2.907e+4	123.863	3249
a_4 [--]	-7.41e+4	0	-3370
a_5 [--]	6.997e+4	0	1305
Thermal Conduction			
λ_{dry} [--]	0.682	0.6	0.06
λ_{mst} [--]	0	0.56	0.56
Heat capacity			
ρ_0 [kg/m³]	1600	230	212
c_0 [J/kgK]	1000	920	1000

4 Dimensions:

A.	Load bearing layer:	0.365 m
B.	Mortar:	0.015 m
C.	Insulation layer:	0.040 m
		Total: 0.420 m

Boundary conditions

1 Thermal and humidity conditions:

Constant values:

Inside: 20°C, 60% Rel.Humidity
Outside : 0°C, 80% Rel.Humidity

2 Pressure conditions: No air pressure difference

3 Surface transfer coefficients:

$\alpha_{e,e}$ = 25 W/m^{2}K, $\alpha_{e,i}$ =8 W/m^{2}K, $\beta_{p,e}$ = 1.838200e-07 s/m, $\beta_{p,i}$ =5.8823e-08 s/m

Initial conditions:

Temperature: T = 25 ºC
Rel. Humidity: ϕ = 60% throughout the whole construction

Output:

1. Simulation time: 60 days

2. Requested output:

The result should be the moisture content profile w(x) and the relative humidity profile ϕ(x) at the end of simulation time.

Output file structure (in ASCII format)

Column 1	Column 2	Column 3
x	*w*	ϕ
(mm)	(kg/m³)	(%)
0.5	x.xxxx	x.xxxx
1.5	…	…
2.5	…	…
…		
8.5	…	…
9.5	…	…
12.5	…	…
17.5	…	…
22.5	…	…
…		
342.5	…	…
347.5	…	…
352.5	…	…
355.5	…	…
356.5	…	…
…		
418.5	…	…
419.5	…	…

All values refer to the time point 60 days after start of simulation. Small x-values correspond to the exterior side of the construction, big values correspond to the interior side. From the load bearing layer only the inner centimetre and the outer centimter have to be given in mm-steps, the rest has to be given in 5mm-steps, to avoid too many data.

3 Names of data files:

The output file should be named as: 'University/Institute'+ Bench5+ '.txt',
e.g. file from Chalmers (CTH):

CTHBench5.txt

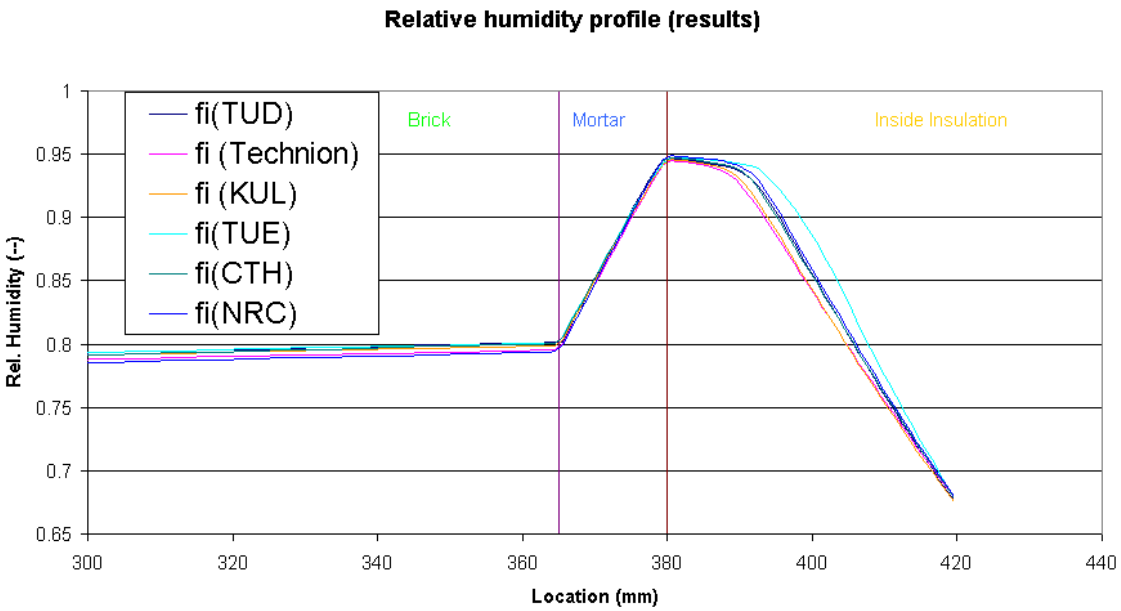


Figure 3.5.1

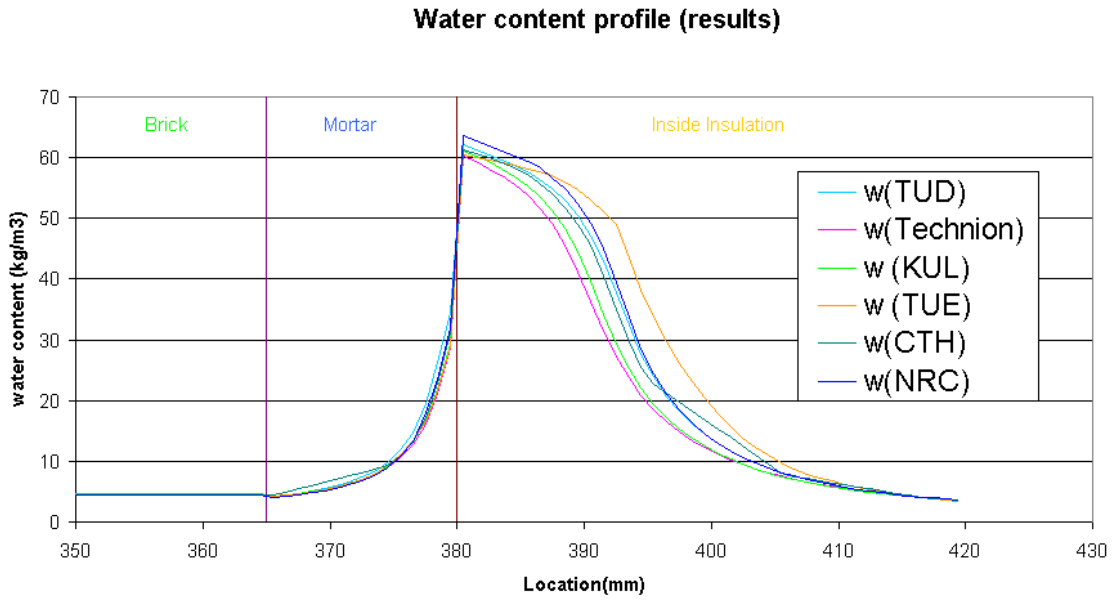


Figure 3.5.2

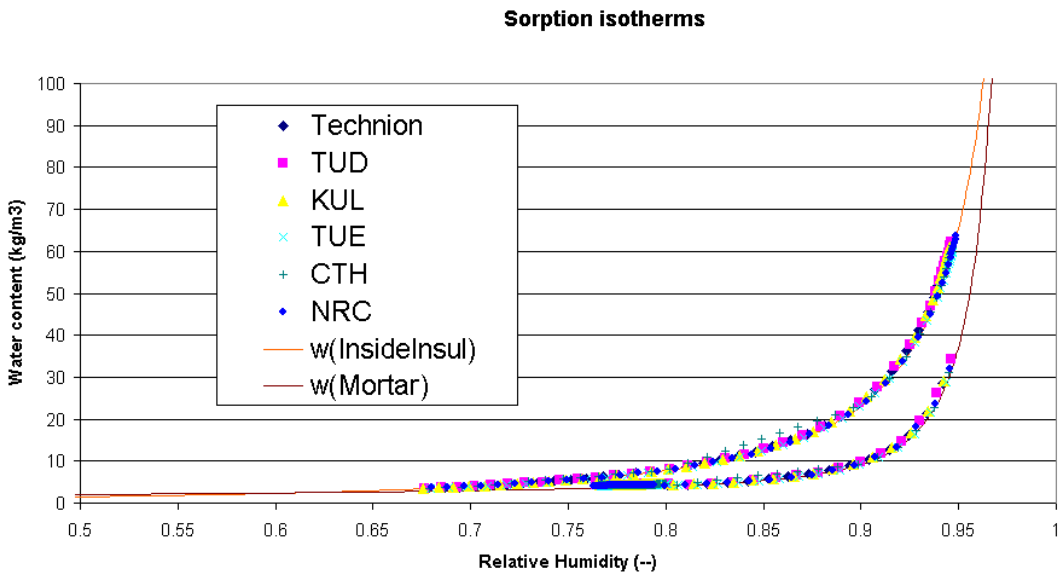


Figure 3.5.3

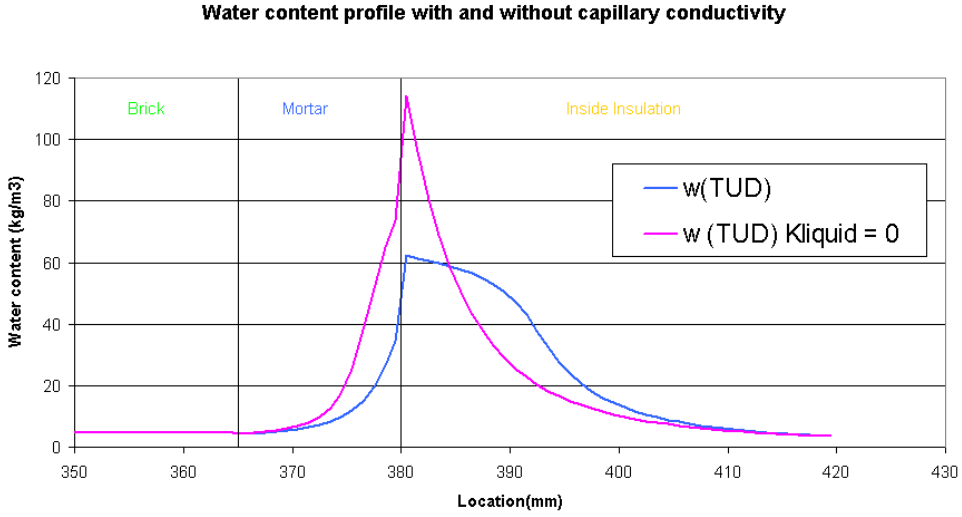


Figure 3.5.4

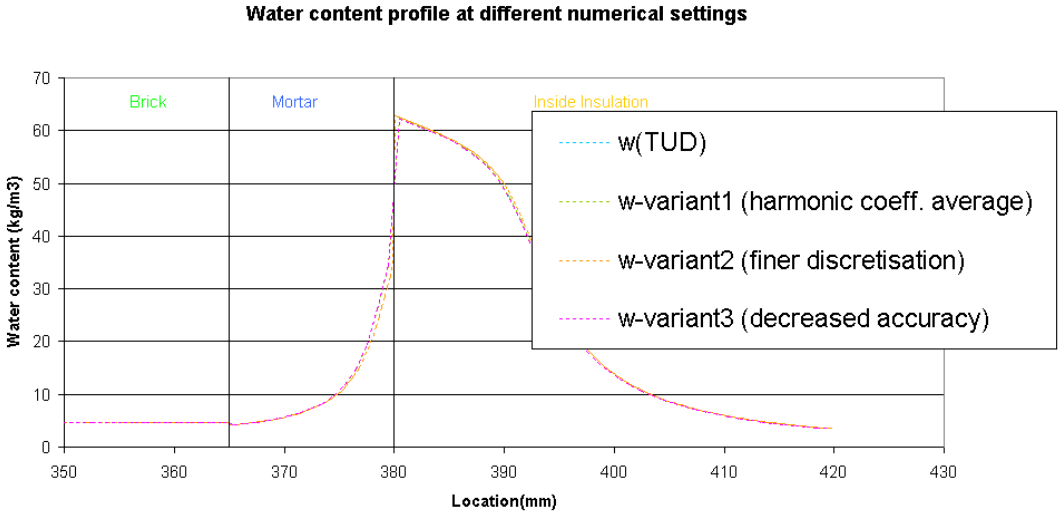


Figure 3.5.5

3.6 Band of acceptance

The transient temperature and moisture conditions are presented in the benchmark one by six different numerical solutions (different softwares), based on the same mathematical model. In order to find a consensus solution, the obtained results are treated statistically.

The t-distribution is used for statistical treatment of data. The choice of t-distribution is based on the fact that the number of observation (obtained results) is low and by this number of observation t-distribution gives a better confidence interval than normal distribution.

This analysis is based on the assumption of random error - produced by the different universities. This can of course be questioned. However, it represents a consistent methodology to analyse the results.

Confidence interval can be calculated by the following equation:

$$\bar{x} - t_p \frac{s_x}{\sqrt{n}} \leq \mu \leq \bar{x} + t_p \frac{s_x}{\sqrt{n}}$$

Where

$\bar{x}$ is the mean value.
s_x is the standard deviation.
μ is the expected value.
n is the number of observation.
p is the total error risk.
t_p is a function of n and p according to table 1

The table below shows t_p as a function of number of observation and total error risk.

Number of observation n	t_p as function of n and p			
	p=20% Confidence grade 100-p=80%	p=10% 100-p=90%	p=1% 100-p=99%	p=0.1% 100-p=99.9%
5	1.5	2.1	4.6	8.6
10	1.4	1.8	3.3	4.8
25	1.3	1.7	2.8	3.7
120	1.29	1.66	2.62	3.3

3.6.1 Results of benchmarks concerning band of acceptance

Moisture and temperature distributions are two important unknown parameters in our calculations. These parameters are time and space dependent and the number of observations for each time and space is about 6. Thus, the standard deviations of these parameters for each time and space are not going to be the same over the time and space. In other words $t_p \frac{s_x}{\sqrt{n}}$ is not the same for each time and space. This lead to different confidence interval for our calculated values. An example of using band of acceptance is shown in figures 3.6.1-3.6.3.

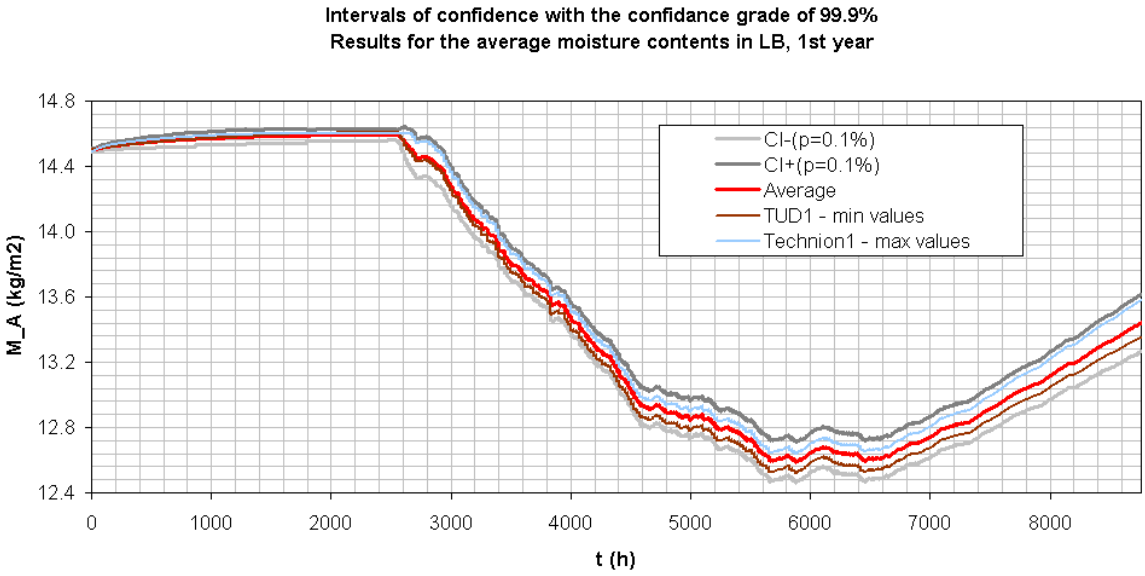


Figure 3.6.1

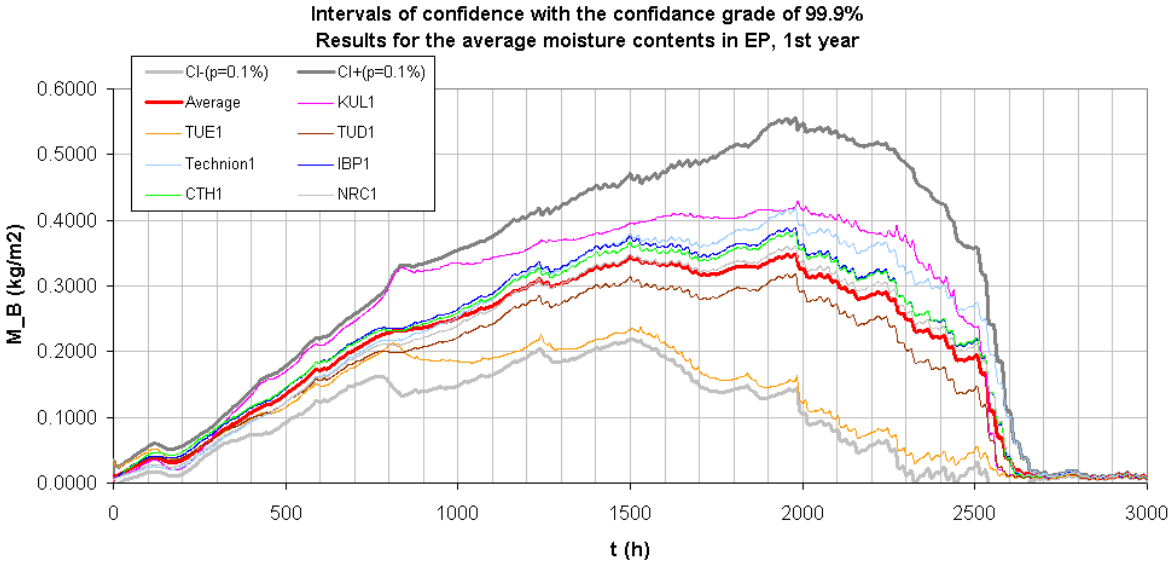


Figure 3.6.2

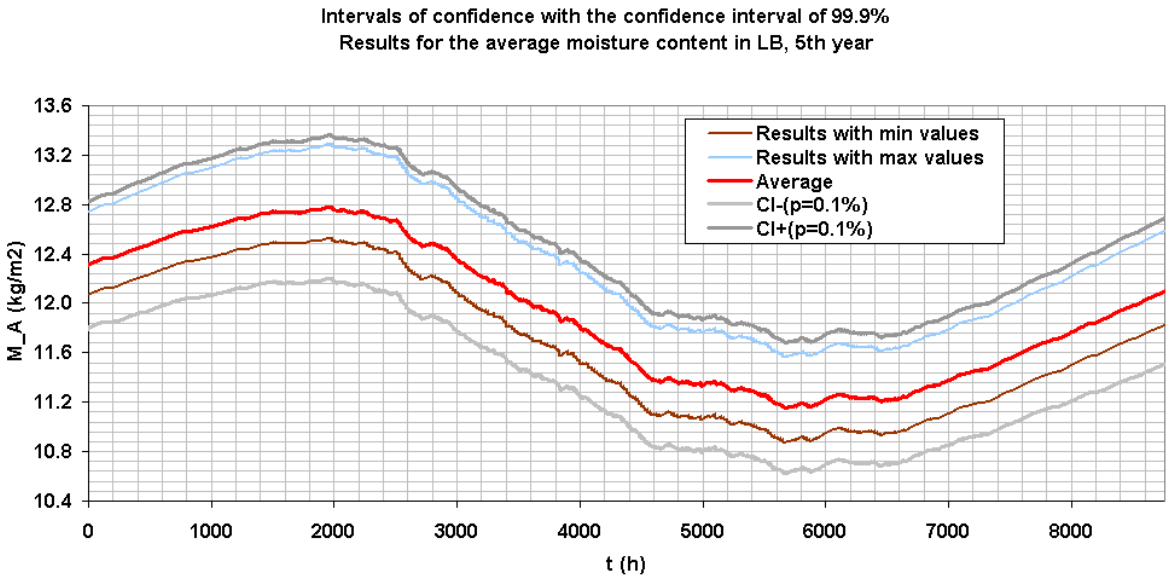


Figure 3.6.3

Appendix: Content of CD

- Final report (The report on your hand.)
- Mathematical modeling
- Benchmarks
    - Benchmark (1):
        - Description of structure
        - Climate data: ClimateBench1.txt
        - Results
            - Graphical presentation
            - Numerical values: excel files
    - Benchmark (2):
        - Description of structure
        - Results
            - Graphical presentation
            - Numerical values: excel files
    - Benchmark (3):
        - Description of structure
        - Results
            - Graphical presentation
            - Numerical values: excel files
    - Benchmark (4):
        - Description of structure
        - Climate data: ClimateBench4.txt
        - Results
            - Graphical presentation
            - Numerical values: excel files
    - Benchmark (5):
        - Description of structure
        - Results
            - Graphical presentation
            - Numerical values: excel files